

Personality-Aware Non-verbal Behavior Generation in Dyadic Interactions

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Amr Amer

angefertigt unter der Leitung von / supervised by
PROF. DR. CHRISTIAN THEOBALT
Dr. Philipp Müller
Dr. Diogo Carbonera Luvizon

begutachtet von / reviewers
PROF. DR. CHRISTIAN THEOBALT
DR. PHILIPP MÜLLER

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Abstract

In dyadic interactions, non-verbal cues are vital for conveying emotions and intentions, significantly affecting communication perception and effectiveness. Previous methods for generating listener behavior responses have focused primarily on limited cues, like facial expressions, and overlooked the influence of personality traits on non-verbal behavior. This thesis introduces our approach to generating non-verbal behavior that captures a broader range of cues, including body language and hand gestures, and integrates the listener’s personality.

Our methodology builds upon existing models by utilizing a transformer-based architecture that integrates detailed motion capture data and personality traits to produce more nuanced and realistic listener responses. This expansion allows for a more comprehensive simulation of dyadic interactions, where the listener’s non-verbal cues are not only reactive but also reflective of inherent personality characteristics.

We evaluate our model on a large face-to-face dyadic dataset using a set of metrics and user study. Our model demonstrates a notable ability to generate distinguishable non-verbal behaviors influenced by Extraversion. Participants were able to clearly distinguish between the introverted and extraverted listener avatars, indicating the model’s effectiveness in capturing personality-driven non-verbal cues. Furthermore, the incorporation of personality traits into the model significantly improved the engagement and interaction quality between the speaker and listener compared to a personality-agnostic baseline.

Our method offers exciting possibilities for more personalized interactions, such as personalized learning, emotionally intelligent virtual agents, and enhanced social connection with telepresence robots. However, there are unresolved challenges that offer directions for future work, including dual response modeling, which involves generating conversational responses for both the speaker and listener to model the full dyadic interaction, and considering factors beyond Extraversion, like Agreeableness and Conscientiousness, for a more comprehensive understanding of non-verbal behavior.

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Chapter 1

Introduction

1.1 Background, and Motivation

Human communication relies heavily on both verbal and non-verbal cues. Non-verbal communication, encompassing elements like eye gaze, facial expressions, body language, and hand gestures, plays a crucial role in conveying emotions, intentions, and underlying messages [15]. These non-verbal signals are particularly significant in dyadic interactions, which involve two individuals, as they facilitate the clear and transparent transmission of verbal information [16, 17]. Studies have shown that non-verbal cues can account for over half of the message being conveyed. This is supported by Mehrabian's communication model [18], which suggests that spoken words or verbal cues define only 7%, tone of voice contributes 38%, while non-verbal cues, particularly body language, make up the remaining 55% [18].

For example, imagine you are giving a presentation and you notice a member of the audience who is actively engaged in your talk. Their body language signals interest and attentiveness, with head nods becoming more pronounced, posture leaning forward, and eyes fixed on you. These non-verbal cues significantly amplify the message their words might convey, demonstrating their interest and attentiveness throughout your talk. This is the power of non-verbal communication; it can subtly, yet powerfully, reinforce the impact of spoken words.

Building upon the important role of non-verbal cues in communication [18], studies have shown that these cues can be further nuanced by individual personality traits [19, 20, 21]. Some research has demonstrated a relationship between the Big Five personality traits, which are the most commonly used model of personality in academic psychology [22], and nonverbal behavior [19, 23]. The Big Five model, a widely accepted personality framework in psychology, proposes five core dimensions: Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism [22]. These traits influence how people express themselves non-verbally, manifesting in aspects like body language and facial expressions [24, 25, 19].

While each of the Big Five traits likely plays a role in shaping non-verbal com-

munication [24, 25, 19], our research focuses on Extraversion for several reasons. Extraversion is characterized by traits like sociability, talkativeness, assertiveness, and enthusiasm [26, 27, 28]. These characteristics are demonstrably linked to outward forms of expression, making Extraversion a natural starting point for investigating the link between personality and non-verbal cues in listener responses. For instance, research suggests that individuals high in Extraversion tend to exhibit behaviors like leaning forward, tilting their head, using energetic gestures, and maintaining frequent eye contact [24, 29, 30, 25, 31]. This highlights the importance of considering personality when generating non-verbal human behaviors to promote effective and natural communication in dyadic interactions.

Modeling human communication in such interactions presents a complex challenge, requiring an understanding of both verbal and non-verbal cues. Non-verbal cues, such as facial expressions, body language, and hand gestures, are vital in shaping listener responses, and enriching verbal messages. Advances in deep learning have led to multi-modal models capable of simulating these interactions [32, 33, 34, 35, 36, 5, 11]. These models have shown promising results in generating listener non-verbal cues that complement the speaker’s message. However, a significant limitation of existing work [34, 35, 36, 5, 11] is the narrow focus on a restricted range of non-verbal cues, which are often limited to facial expressions, or head nods indicating agreement or disagreement. Additionally, most current models do not consider personality’s impact on non-verbal behavior.

In this thesis, we aim to delve deeper into the dynamics of dyadic communication, with a particular focus on the role of personality in shaping non-verbal behavior. Our goal is to expand upon existing research by incorporating a broader range of non-verbal cues, encompassing facial expressions, body language, and hand gestures, while also accounting for individual differences in personality traits. By doing so, we hope to develop a more comprehensive understanding of the complex interplay between personality and non-verbal communication in dyadic interactions.

1.2 Research Objective and Questions

Our research objective is to develop a comprehensive model for generating listener non-verbal behaviors in dyadic conversations that reflects the listener’s personality traits, and responds to the speaker’s input. The model will incorporate a wide range of non-verbal cues, including facial expressions, body posture, and hand gestures to capture the subtle complexities of one-on-one human communication. By doing so, we hope to enhance the realism and engagement of generated listener responses, thereby improving upon existing methods [11, 34, 35, 36, 5].

To achieve this objective, the research addresses the following questions:

1. Is it feasible to build a model that generates non-verbal behaviors that encompass the entire upper body, including facial expressions, arms, hands, and body posture,

and that are reflective of the listener’s personality trait?

2. Can the incorporation of listener personality traits into the model improve the realism, and engagement of listener responses to the speaker?

In the conclusion chapter of this thesis, we will revisit these research questions, and provide answers based on the findings of our studies and results.

1.3 Thesis Structure

This thesis is structured as follows:

Chapter 1: Introduction

- Introduced the research problem, highlighting the significance of personality in non-verbal communication, and outlined the research objectives and questions addressed in our work.

Chapter 2: Related Work

- Surveys existing research on the connection between personality and non-verbal behavior.
- Reviews current state-of-the-art methods for non-verbal behavior generation.

Chapter 3: Background

- Presents essential background knowledge necessary to understand our proposed approach.

Chapter 4: Methodology

- Details our data preprocessing pipeline, and feature extraction techniques.
- Introduces our proposed multimodal personality-aware model for listener response generation.

Chapter 5: Experiments and Evaluation

- Describes the UDIVA dataset used for model training and testing.
- Explains the experimental setup, evaluation metrics, and implementation details.
- Presents the results obtained from our model and user studies, comparing them to baselines.

Chapter 6: Discussion and Future Work

- Summarizes the key findings of the thesis.
- Discusses the limitations of our work, and suggests potential avenues for future research, such as incorporating additional personality traits.

Chapter 7: Conclusion

- Recapitulates the most significant points, and contributions of the thesis.

Chapter 2

Related Work

In this chapter, we delve into the complex interplay between personality traits, nonverbal communication, and the modeling techniques used for predicting nonverbal behavior in dyadic interactions. Our exploration begins with an in-depth look at the profound influence of personality traits, with a particular focus on extraversion, and how they shape nonverbal communication patterns.

The focus subsequently shifts to the challenges and advancements in modeling nonverbal behavior prediction in dyadic interactions. We trace the journey from early rule-based methods to more advanced techniques, including data-driven approaches and transformer-based models, highlighting the evolution and innovation in this field.

In concluding this chapter, we acknowledge the limitations of 2D key points in capturing the full spectrum of nonverbal behaviors in dyadic interactions. Looking ahead, we pivot our attention to 3D human modeling technologies. These advanced methods promise a more nuanced and comprehensive representation, offering a deeper insight into effectively capturing nonverbal cues.

2.1 Interplay of Personality and Human Behavior

The relationship between personality traits and Human behavior is a dynamic and complex one. Personality traits shape our thoughts, emotions, and behaviors, which in turn influence our interactions with the world around us. This ongoing interplay between personality and behavior has significant implications for our relationships, our work, and our overall well-being.

This relationship between personality traits and behavior extends to both verbal and nonverbal communication. Verbal behavior refers to the words that we use to express ourselves, while nonverbal behavior encompasses all non-linguistic cues that we convey, such as our facial expressions, body language including body postures and gestures.

A significant finding in psychology is the relationship between personality traits and nonverbal behavior. Studies have revealed that individuals with different personality traits exhibit distinct patterns of nonverbal behavior. For instance, individuals with high Extraversion tend to display behaviors like leaning forward, head tilting, energetic gestures, and frequent eye contact, according to various research sources such as [24, 29, 30, 25, 31].

2.1.1 Introduction to Personality and Nonverbal Behavior

Research has consistently shown that there is a strong correlation between personality and nonverbal behavior[37, 38]. This means that people who are high in certain personality traits such as extraversion and agreeableness tend to exhibit certain nonverbal behaviors such as more interactive and physically active , while people who are low in those traits tend to exhibit different nonverbal behaviors such as less interactive and physically conservative or Inactive.

Ambady and Rosenthal [39] demonstrated that people can make accurate judgments of personality traits, such as extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience, from just a few minutes of observation. This phenomenon, referred to as thin-slices of behavior, highlights the ability of nonverbal cues like facial expressions, and body language to provide insights into an individual’s underlying personality.

Personality, and its various facets, can be encapsulated by the Big Five Personality Traits. Each of these traits corresponds to unique patterns of non-verbal behavior. These traits and their related non-verbal behaviors will be explored in detail in the upcoming sections.

2.1.2 The Big Five Personality Traits (OCEAN)

The big five personality traits, also known as the five-factor model (FFM) is a widely recognized framework for understanding individual personality. This model introduced by Costa Jr and McCrae [14], proposes that personality can be effectively categorized into five broad dimensions: neuroticism, extraversion, openness to experience, agreeableness, and conscientiousness. These dimensions, often summarized using the acronym OCEAN, capture the fundamental aspects of human personality.

The Big Five Personality Traits are further divided into specific facets that represent

the different aspects of each trait. Each trait has two opposing ends, and the facets describe both the high and low ends of each trait. Table 2.1 provides a summary of the Big Five Personality Traits and their associated facets.[14]

Personality traits	Facets
Extraversion (E)	Talkative, a joiner, physically active, affectionate, passionate, fun-loving
Introversion	Reserved, seeking solitude (a loner), physically passive, quiet, sober, unfeeling
Open to new experiences (O)	Imagination, creativity, originality, prefer variety, curiosity, liberal
Traditionalist	Down to Earth, uncreative, conventional, uncurious, prefer routines, conservative
Conscientiousness(C)	Conscientious, hard-working, ambitious, well organized, persevering, punctual
Careless	Quitting, negligent, lazy, disorganized, aimless, late
Agreeableness (A)	Softhearted, trusting, generous, acquiescent, lenient, good-natured
Self-centered	Suspicious, ruthless, stingy, antagonistic, critical, irritable
Neuroticism (N)	worrying, temperamental, self-pitying, self-conscious, emotional, vulnerable
Emotionally stable	Calm, even-tempered, self-satisfied, comfortable, unemotional, hardy

Table 2.1: Personality traits and facets. Taken from [14]

The Big Five Traits shape each individual’s personality, but one trait often dominates, influencing their behavior and perspective. Research suggests that individuals with higher levels of extraversion and emotional stability exhibit more noticeable movement patterns, which can be readily detected by human observers.[40]

2.1.3 Extraversion: Influence on Listener Behavior Dynamics

Based on existing research on personality traits [41, 42, 43], we can infer correlations between some nonverbal communication cues and each personality dimension. Among these dimensions, extraversion has been found to have notable associations with nonverbal behaviors [41, 24, 29, 30, 25, 31].

- **Gazing behavior:** Individuals scoring high on extraversion tend to reciprocate gazes and maintain prolonged eye contact during conversations, indicating engagement and interest.[41]
- **Proximity behavior:** Extraverts are having more frequent and more rapid body movements [41, 24, 29, 30, 25, 31].
- **Response to unexpected proximity:** When approached unexpectedly, individuals high on extraversion are less likely to step back, indicating a comfort level with close proximity [41, 24, 29, 30, 25, 31].
- **Expressive body language:** Extraverts often exhibit more animated and expressive body language, using gestures, facial expressions, and posture to amplify their communication. Their movements tend to be more energetic, dynamic, and varied compared to those of introverts [41, 24, 29, 30, 25, 31].

2.2 Nonverbal Behavior Prediction

In this section, we focus on the challenge of modeling nonverbal feedback in dyadic interactions. This task is complex due to the unpredictable nature of listener responses, and the multi-modal aspects of speakers which simultaneously engage in verbal communication through both speech and nonverbal expression via facial and body gestures. As we have explored how personality traits of both the speaker and the listener influence nonverbal reactions in last section. This understanding leads us to the need for a model that integrates personality factors, as previously detailed in Section 2.1 discussing how these traits impact non-verbal behaviors like facial expressions and body motions.

We start with early rule-based approaches that employed basic assumptions to predict nonverbal behavior, but their scope and realism were limited [44, 45, 46, 47]. Progressing to data-driven methods, we discuss the advancements made by generative adversarial networks (GANs), and comparative models in synthesizing nonverbal motions. However, they usually face challenges in capturing the complete range of human nonverbal cues such as limited facial and body gestures [48, 49, 3, 2].

Next, we consider predictive methods like auto-encoder based methods, which include a wider variety of cues but still face challenges in producing realistic motions [50, 51, 5, 35]. The latest transformer-based methods, particularly learning to listen by Ng et al. [11], stands out for their ability to generate realistic motions through understanding long-range dependencies in interactions. However, despite the importance of personality in shaping nonverbal behaviour, these approaches are completely agnostic to personality traits.

This section provides a comprehensive overview of the evolution and current challenges in predicting nonverbal behavior in dyadic conversation synthesis, highlighting the need for methods that capture the complex relationship between personality traits and nonverbal interactional communication.

2.2.1 Early Rule-based Approaches

Early rule-based approaches focused on analyzing features from various modalities to generate appropriate nonverbal responses in dyadic interactions. These initial studies relied primarily on manual or rule-based methods to analyze video-recorded data. Despite these challenges, the early stages of this research significantly advanced our understanding of nonverbal communication in personal interactions. Chu et al. [1] and Cohn and Tronick [52] were particularly influential in highlighting how a listener’s nonverbal behavior tends to synchronize with the speaker’s actions in mother infant face to face interactions. They showed some experimental to explore the concept of interactional synchrony introduced by Condon and Ogston [53], revealing how listeners’ nonverbal behaviors rhythmically align with those of speakers. Figure 2-1 demonstrates

[1] approach to identifying synchrony in a mother-infant interaction, highlighting the mutual engagement of smiles between the two. However, the method's reliance only on overlapping face video frames and multi-dimensional signals like histograms as a measure of synchrony could pose limitations in diverse interaction contexts.



Figure 2-1: Automatically discovers dyadic synchronises from multi-dimensional signals. Red bold boxes indicate the engagement in mutual smiles between the infant and the mother. The gray thin box indicate a randomly picked moment, showing an event without synchrony. Adapted from Chu et al. [1]

Kendon [54] and Ramseyer and Tschacher [55] further extended those studies approaches [1, 53] based on interactional synchrony concept [53] by manually transcribing a broader range of nonverbal behaviors in video recordings, including head movements and upper body gestures, demonstrating the intricate interplay of non-verbal cues in dyadic interactions. In one of studies by Kodama et al. [56], employed rule-based methods to analyze video recordings of a psychotherapy session. This study investigated the nonverbal coordination between psychotherapists and clients during counseling sessions. Using a combination of quantitative methods (like video analysis and nonlinear time series analysis of body movements) and qualitative approaches (including direct observation and therapist interviews). Key findings indicated high levels of bodily coordination during critical therapy stages like rapport building, clinical interventions, and session closure. These identified synchrony patterns highlight the importance of physical synchrony in reflecting the therapy's effectiveness and the client-therapist relationship.

2.2.1.1 Limitations

Early rule-based approaches to analyzing motion patterns in dyadic conversations faced significant limitations that hindered their ability to capture complexities of nonverbal communication.

First, those methods often employed simplistic assumptions for motion generation, often focusing on isolated nonverbal cues and limited variety of captured gestures such as head nods or facial expressions as we have seen in those studies by [1, 52].

Second, early rule-based methods utilized customized or simple analysis techniques [56] that were not scalable to large datasets. This restricted their applicability to real-world interactions involving diverse nonverbal cues. For example, [56] used a rule-based method to analyze video recordings of psychotherapy sessions, but their method could not handle large datasets and required manual coding of nonverbal behaviors.

In conclusion, early rule-based methods laid the foundation for understanding nonverbal communication in dyadic interactions [53, 54]. However, their limitations in terms of data reliance, analysis complexity, and generalizability have prompted the development of more sophisticated machine learning-based approaches. Machine learning models or data driven approaches can learn complex patterns from large datasets, enabling them to capture the natural synchrony in dyadic conversations with greater accuracy and detail.

2.2.2 Data-Driven Approaches

2.2.2.1 GANs and Comparative Models

Prior data-driven approaches have primarily concentrated on predicting the 2D facial expressions of one individual in a conversation by analyzing the motion patterns of their partner in a set of n dyadic interactions. Huang and Khan [2] introduced a two-stage generative adversarial network (GAN) [57, 58] architecture for synthesizing facial expressions in interviewer-interviewee dyadic interactions, as shown in Figure 2-2. The first stage generates expressive 2D face sketches of the interviewer, conditioned on the facial expressions of the interviewee. These face sketches, which are sets of 68 landmark 2D points that depict components of human faces, such as eyes, eyebrows, nose, mouth, and face out-contour, are then used as input to the second stage. The second stage generates complete face images conditioned on the facial expressions encoded in the face sketches from the first stage.

Despite generating appropriate facial expressions for individual frames, the proposed approach by Huang et al. [2] lacks in maintaining temporal coherence over time, leading to unnatural and discontinuous facial movements. This shortfall arises because GANs can suffer from mode collapse [59, 60, 61], while proficient at generating

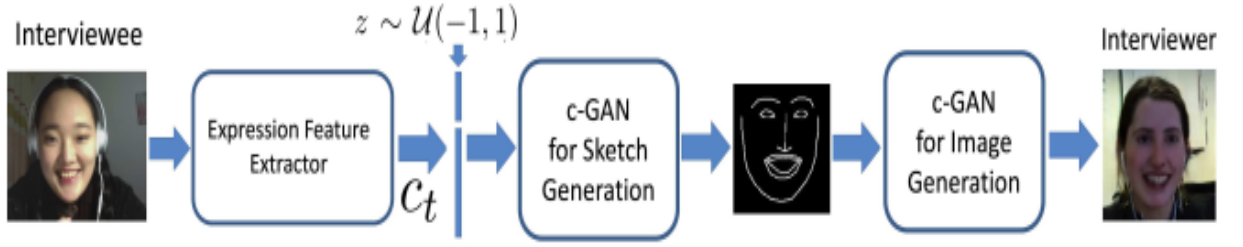


Figure 2-2: DyadGAN is composed of two stages of GANs, one to generate sketches and the other one to generate facial expressions of the interviewer. The inputs are facial expression features extracted from images of interviewees that serve as the conditioning vector c_t . Adapted from Huang and Khan [2]

high-quality individual images, often struggle with temporal dependencies crucial for smooth transitions in video sequences. Moreover, it struggles to effectively capture crucial nonverbal cues such as head position and orientation, pivotal for portraying behaviors like head tilts, nods, and shakes. This limitation stems from GANs’ focus on static images, which does not naturally extend to the dynamic aspects of nonverbal communication that evolve over time.

In an effort to address these limitations, Nojavanasghari et al. [3] and Sadoughi et al. [62] innovated upon previous methodologies by incorporating a Bidirectional Long Short-Term Memory (BLSTM) [63]. This enhancement is motivated by the BLSTM’s capability to process sequences in both forward and backward directions, allowing for a more comprehensive understanding of temporal dynamics and resulting in more coherent generation of nonverbal behaviors over time.

However, despite their advancements in capturing a wider array of nonverbal cues, the implementation presented by Nojavanasghari et al. [3] in their results (Figure 2-3) reveals persistent challenges with achieving seamless transitions and maintaining natural sequences over time. The observed discontinuities and awkward transitions may be attributed to the inherent complexity of modeling human nonverbal communication, which involves subtle and continuous changes that are difficult to replicate accurately with current machine learning models. Furthermore, their approach, confined to the generation of 2D facial keypoints, falls short in fully encapsulating the depth and intricacy of nonverbal behaviors. This limitation highlights the model’s inability to capture the three-dimensional aspects of facial expressions and head movements, underscoring the need for models that can generate more complex, three-dimensional nonverbal cues to achieve a higher degree of realism.

More recently, researchers have explored modeling a speaker’s fine-grained motion generation conditioned on additional information such as audio [4], text [34], or both audio and text [64]. Despite the advancements in incorporating those additional information, it is important to note that these methods primarily focus on modeling monadic conversational settings, aiming to generate speaker motion that directly

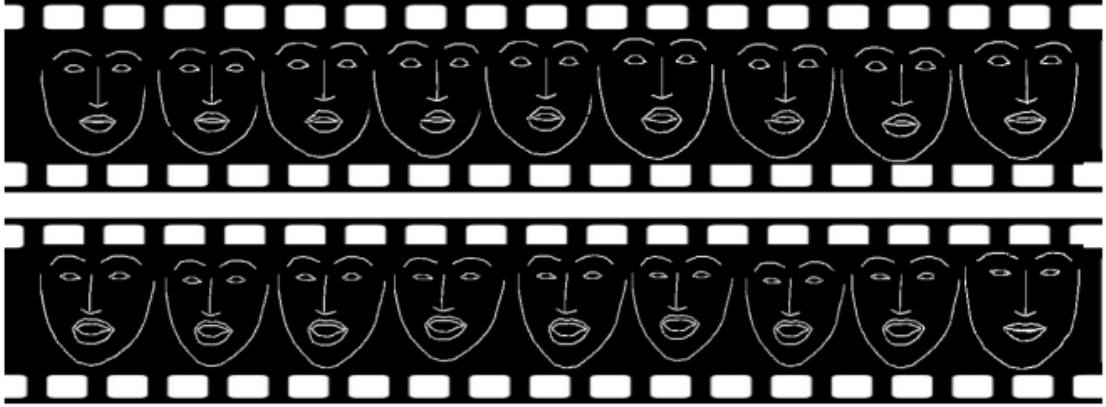


Figure 2-3: Examples of generated 2D facial expressions. Adapted from Nojavanasghari et al. [3]

corresponds to the input signals. Moreover, some of these approaches only synthesize speaker gestures without considering the listener’s input. As a result, these methods are unable to fully capture the complexities of cross-person, dyadic interactions by predicting the listener’s nonverbal behaviors.

In addition to their focus on monadic interactions, several recent works also face limitations in their use of 2D keypoints representations as shown in Figure 2-4. While these representations offer convenience, they lack the expressiveness to capture the full range of nonverbal cues, such as subtle facial expressions and detailed body motions. For instance, papers [4, 64] represent the speaker’s pose using a temporal stack of 2D skeletal keypoints, obtained using OpenPose [6]. This limited representation fails to capture the full range of body motions, potentially hindering the ability to accurately model nonverbal communication. Similarly, paper [34] restricts its analysis to facial expressions, using only a set of 68 facial keypoints [65]. This approach may overlook subtle facial cues and expressions, which are crucial for understanding emotional states and intentions. The reliance on 2D keypoints representations restricts the expressiveness and nuance of nonverbal behaviors captured by these methods, limiting their ability to fully capture the complexities of dyadic interactions. Having addressed the limitations of 2D keypoints representations, we will turn to the realm of 3D face and human modeling methods, which offer a more comprehensive and expressive approach to capturing nonverbal cues. Which will be explored in detail in later on section 3.1

2.2.2.2 Auto-Encoder Based Methods

Auto-encoders ability to model complex probability distributions and incorporate multi-modal input, including audio, text, and facial expressions, allows them to synthesize a wider range of nonverbal cues beyond the limited set often captured in existing methods



Figure 2-4: Examples of corresponding input audio, and synthesized 2D facial and body key-points overlaid on video frames from In-the-wild conversational dataset. Adapted from [4]

[2, 3, 34], enabling the generation of more diverse, context-aware, and contextually appropriate nonverbal behaviors. Feng et al. [5] employed a variational autoencoder (VAE) to encode the model’s input 2D key points tracked on the interlocutor’s face. By analyzing the history of encoded facial expressions from a corpus of videos depicting pairs of individuals conversing, their proposed model successfully predicted both high-frequency (e.g., eye blinks, mouth movements) and low-frequency (e.g., head nods) changes in the listener’s facial expressions at each time interval. The use of a VAE for encoding facial key points as shown in Figure 2-5 enhanced the model’s generalization ability and preserved valuable nuanced expressions that can be overlooked when expressions are categorized into discrete emotional states [34].



Figure 2-5: Illustration of the effect of changing values in 10 of the VAE dimensions from -4 to 4. Notice that each dimension affects a different aspect of the face, e.g., face orientation, opening the mouth, opening the eyes, smiling, etc. Adapted from [5]

In contrast to previous methods [4, 34, 64] that focused on modeling monadic conversations, focusing on the speaker’s facial motions and using 2D representations to encode facial gestures, Zhou et al. [66], Jonell et al. [35, 36] introduced a new approach that utilizes animated 3D face avatars. This approach employs FLAME [67], a novel parameterization technique that represents facial expressions, shapes, and head rotations in a low-dimensional Principal Component Analysis (PCA) parameter space [68]. These parameterized representations are then mapped onto 3D meshes, enabling the generation of realistic facial animations. Moreover, these methods extend beyond monadic interactions and encompass cross-person, dyadic interactions by predicting the listener’s 3D facial motion. Jonell et al. [35] proposed an autoencoder-based model that incorporates multi-modal inputs of 3D speaker and listener motion, and speaker speech audio to generate diverse nonverbal facial cues. The model leverages a normalizing flow [69] to transform Gaussian driving noise into a distribution of facial expressions, while leveraging auto-regressive conditioning to generate smooth facial motion. The model is further conditioned on the avatar’s speech and the interlocutor’s speech and facial features to ensure consistency and appropriate behavior. The model’s reliance on normalizing flows to generate facial expressions hinders its ability to capture long-range dependencies in human nonverbal behavior. Normalizing flows, by their nature, operate by transforming simple distributions into more complex ones, effectively smoothing out the input data. While this approach ensures smooth and natural-looking motions, it sacrifices the ability to capture the subtle nuances and intricate relationships between gestures that characterize human nonverbal communication.

2.2.2.3 Limitations of Data-Driven Approaches

Data-driven approaches for predicting nonverbal behavior in dyadic interactions face several limitations that hinder their ability to accurately capture the complexities of human communication. These limitations include:

- **Inability to capture long-range dependencies:** Traditional data-driven methods, such as GANs and autoencoders, often struggle to capture the intricate relationships between nonverbal cues and the broader context of the interaction. This is because these methods typically process the input data sequentially, without considering the underlying temporal structure and long-range dependencies.

- **Relying on 2D keypoints representations:** Several data-driven approaches utilize 2D keypoints to represent nonverbal cues. While 2D keypoints offer a convenient representation for tracking facial landmarks and body joints, they lack the expressiveness to capture the full range of nonverbal cues, such as subtle facial expressions, nuanced body movements, and intricate interactions between upper and lower body movements.
- **Limited expressiveness and nuance:** The reliance on 2D keypoints representations restricts the expressiveness and nuance of nonverbal behaviors captured by these methods. 2D keypoints fail to capture the subtle details and subtle cues that are crucial for understanding emotional states, intentions, and the dynamics of dyadic interactions.
- **Overlooking temporal coherence and smoothness:** Some data-driven approaches struggle to generate temporally coherent and smooth nonverbal motions. This is evident in the presence of unnatural transitions and discontinuities between consecutive frames.

2.2.3 Transformer-Based Methods: A Promising Alternative

Transformer-based methods have emerged as a promising alternative to traditional data-driven approaches for predicting nonverbal behavior in dyadic interactions. These methods offer several advantages over traditional approaches, including their ability to capture long-range dependencies, model dyadic interactions effectively, capture 3D nonverbal cues, and integrate audio and text modalities. Additionally, Transformer-based methods can generate diverse, context-aware, and natural nonverbal cues. These capabilities make Transformer-based methods a promising tool for developing more natural, engaging, and empathetic human-computer interactions [10, 11, 46, 70].

Recent research has demonstrated the effectiveness of Transformers in generating a wide range of motions for various applications while effectively capturing intricate long-range relationships. For example, Li et al. [10] proposed FACT, a Full-Attention Cross-modal Transformer network, capable of generating 3D dance motions in sync with music. This model utilizes a deep cross-modal transformer block with full-attention to capture complex relationships between music and motion, enabling it to predict multiple future motions and produce long, coherent dance sequences.

Another example of Transformers effectiveness in generating motions is the ACTOR model proposed by Petrovich et al. [71], a Transformer-based architecture trained to generate diverse and realistic human motion sequences conditioned on specific actions like “Throw this object”. This model utilizes a VAE (variational autoencoder) to learn a latent representation of human motions, enabling it to generate a wide range of motion sequences tailored to the given action. The Transformer’s ability to capture long-range dependencies within motion sequences ensures temporal coherence and naturalness. Additionally, this model parameterizes the human body using SMPL [7],

a 3D human modeling technique that as discussed in earlier sections, can capture more nuances of body poses than 2D human motion representations. For a more detailed discussion of SMPL, please refer to the next section 3.1.

In the context of modeling nonverbal behavior in dyadic interactions, the paper "Learning to Listen" by Ng et al. [11] stands out as a significant contribution. This study addresses the complexities of dyadic interactions by focusing on the listener's facial motion in response to a speaker, a critical aspect that is often overlooked in traditional models such as GANs and autoencoders. The key feature of this method lies in its innovative use of a transformer-based predictor adapted from Li et al. [10], which effectively captures long-range dependencies, a crucial ability considering the nuanced and evolving nature of dyadic interactions. Additionally, the incorporation of a cross-attention mechanism between motion and audio inputs using cross-model transformers [72] is noteworthy, as it enables the method to gain a comprehensive understanding of nonverbal communication, recognizing that verbal and nonverbal cues are closely correlated in dyadic settings. Furthermore, the model's ability to generate realistic 3D listener facial motion synchronized with the speaker's expressions is a significant breakthrough, as this synchronization is essential for creating natural and believable interactions. The non-deterministic aspect of the model is also crucial, as it acknowledges the inherent unpredictability and variability in human interactions, unlike deterministic models that might oversimplify or fail to capture the full range of human expressions.

2.2.4 Summary

In summarizing this section 2.2, we have delved into the intricate task of modeling non-verbal feedback in dyadic interactions, a challenge compounded by the unpredictable and multimodal nature of human communication. Our exploration began with the influence of personality traits on nonverbal reactions, underscoring the necessity of integrating these factors into predictive models.

We reviewed the progression from early rule-based approaches with limited realism to more advanced data-driven methods, including GANs and auto-encoders, each improving upon the synthesis of nonverbal motions but still facing limitations in fully capturing human expressiveness. Notably, transformer-based methods, exemplified by Learning to Listen by Ng et al. [11], have shown promise in generating realistic motions by understanding interaction dynamics, yet they too have gaps, particularly in integrating personality traits into nonverbal behavior prediction.

Chapter 3

Background

In this chapter, we delve into the background of our work, beginning with an exploration of 3D human modeling methods that focus on monocular 3D human mesh recovery from images, also known as 3D human pose and shape estimation. We will then discuss relevant models such as FACT by Li et al. [10], and Learning to Listen by Ng et al. [11], which provide a foundation for our work, and share connections with our approach. In summary, we will discuss the limitations of the current state of research, and identify potential areas for improvement. This examination will provide valuable insights for refining our approach.

3.1 3D Human Modeling

While simple abstractions of the human body, such as stick Figure 3-1(a), can provide a basic representation of movement, they fall shortly in capturing the intricacies of nonverbal communication in dyadic interactions. In these interactions, individuals engage in a complex interplay of body language, facial expressions, and subtle movements, all of which contribute to the conveyance of emotions, intentions, and interpersonal dynamics. Therefore, for accurate modeling of nonverbal behavior in dyadic interactions, a more sophisticated representation of the human body is required. This is where 3D human modeling techniques come into play, offering a detailed, and nuanced representation of both body pose and shape, enabling the capture of these subtle cues, that play such a vital role in dyadic communication.

Given the availability of monocular videos of dyadic interactions, this section of the related work will primarily explore approaches to monocular 3D human mesh recovery, also known as 3D human pose and shape estimation. Recovering 3D human mesh from monocular images is a challenging task due to several inherent ambiguities, including the limitations of lifting 2D observations onto the 3D space, the complex kinematic structures of the human body, the intricate interactions with the environment, and the scarcity of annotated 3D data. This section will provide an overview of some of the methods in Figure 3-2 employed in this context without delving into extensive

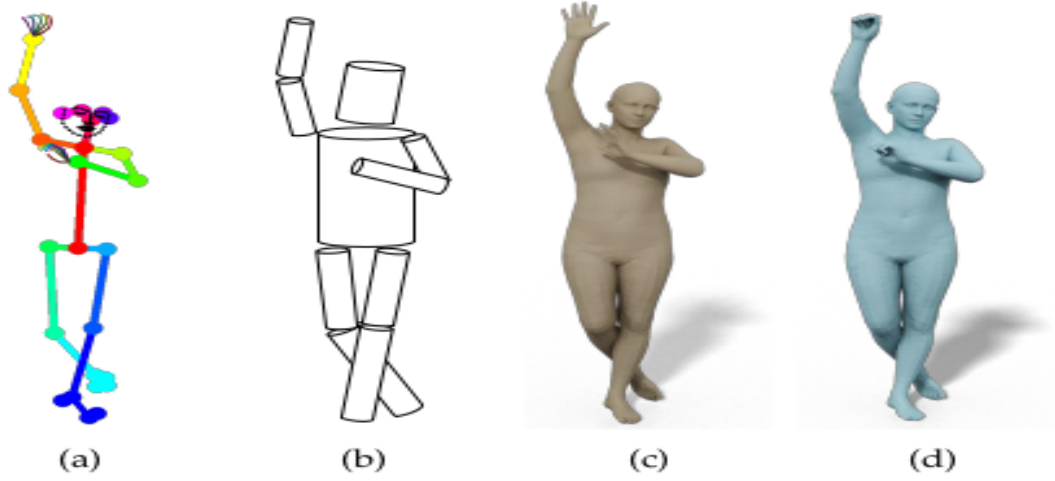


Figure 3-1: Comparison of 2D and 3D human models representing same human pose. (a) 2D skeleton model [6], representing body, face, and hands 2D key points, (b) a cylindrical body mode, (c) SMPL 3D model [7], (d) SMPL-X 3D Model [8]. This Figure is adapted from [9]

technical details, highlighting their limitations, and introduce widely used 3D human models like SMPL Loper et al. [7], SMPL-X Pavlakos et al. [8], and lastly PIXIE Feng et al. [12], which offer well-aligned 3D models.

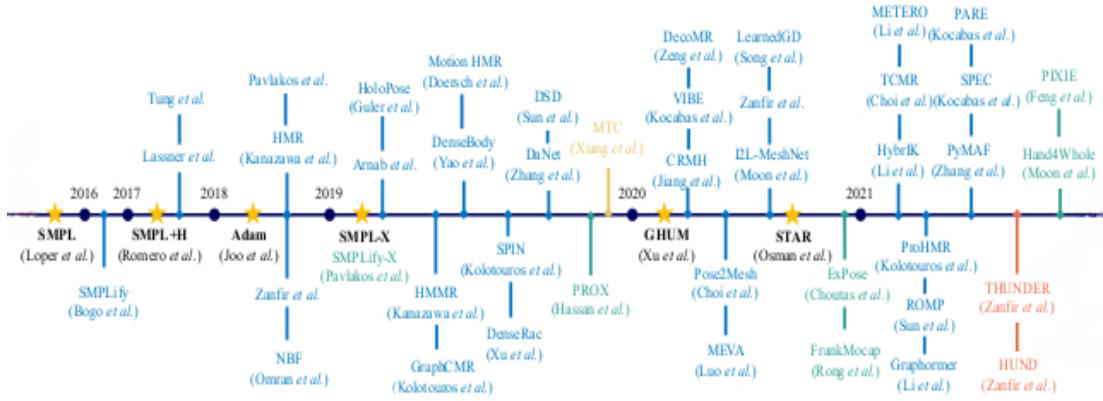


Figure 3-2: overview of 3D human mesh recovery methods. Adapted from [9]

Early human body modeling techniques relied on the manipulation of basic geometric primitives, as depicted in Figure 3-1(b). These methods provided limited details, but advancements in 3D full-body scanning led to more sophisticated approaches, that capture intricate body poses and shapes. Substantial research has been conducted to create highly realistic human body models from 3D scanning data, such as the CAESAR dataset [73]. Among these models, SMPL [7] in Figure 3-1(c) , SMPL-X [8]

in Figure 3-1(d) , and PIXIE [12] stands out for its ability to generate high-fidelity 3D human models. We will discuss those methods in the following sections.

3.1.1 SMPL: A Skinned Multi-Person Linear Model

SMPL is a vertex-based linear model that represents humans in realistic poses, and is highly popular among researchers. It uses two sets of parameters to define the body: shape and pose. These parameters effectively manage the body’s pose deformation, and shape variation, making SMPL a versatile tool for human body modeling.

The pose parameters represented as a series of rotation vectors, are defined by a standard skeletal rig consisting of 23 joints. Each rotation vector denotes the relative rotation of a specific body part with respect to its parent in the hierarchical structure of the kinematic tree. The first rotation vector in this series corresponds to the root orientation, which sets the overall orientation of the model.

The shape parameters are coefficients that represent the most significant variations in body shape, determined through a technique called principal component analysis (PCA) [74, 75, 76]. These coefficients capture the essence of body shape variations in a compact, low-dimensional space. The SMPL model can be understood as a function that takes these shape parameters, along with pose parameters, and maps them to a detailed mesh consisting of 6,890 vertices. This process is structured as an additive model in the vertex space, meaning that the final shape is constructed by adding these calculated deformations to a base mesh.

The SMPL model family, including the FLAME face model [67], and MANO hand model [77], has seen significant adoption. These models utilize linear blend skinning combined with shape, and pose blend shapes. Despite their widespread application, certain limitations of the SMPL framework are noteworthy:

- The global blend shapes in SMPL may capture unwanted long-range correlations, leading to artifacts in non-local deformation.
- SMPL does not consider the relationship between body shape, and pose-dependent shape deformation, which is crucial for realistic modeling.
- The framework’s reliance on a linear PCA subspace for soft-tissue deformations is inadequate for depicting highly non-linear deformations, restricting its ability to portray complex body movements accurately.

Pavlakos et al. [8] introduced SMPL-X, a comprehensive model that jointly represents the human body, face, and hands. They expanded SMPL by integrating the FLAME head model [67], and the MANO hand model [77]. SMPL-X is particularly relevant to our study, as facial expressions, and body poses play crucial roles in nonverbal communication during dyadic interactions. We will delve into the specifics of SMPL-X in the following section.

3.1.2 SMPL-X: A Comprehensive Human Model

SMPL-X is a comprehensive model with several parameters representing the body, hand, and face. The parameterization includes:

- **Rotational Parameters:** 75 parameters for global rotation, and specific poses of the body, head, and jaw.
- **Hand Poses:** Two options are available:
 - 24 low-dimensional PCA coefficients.
 - 90 rotational parameters.
- **Body Shape:** 10 parameters dedicated to defining the body shape.
- **Facial Expressions:** 10 parameters specifically for facial expressions details.

The SMPL-X model integrates the MANO hand model [77], and the FLAME head model [67] to create an expressive representation of the human body. However, it presents certain limitations:

- **Isolation of Components:** The MANO, and FLAME models are learned independently of the body. This approach does not fully capture the interconnected motion, and degrees of freedom of the head, and hands in natural body movements.
- **Artifacts at Boundaries:** Fusing parameters from the MANO, and FLAME models, which are learned separately, with the body model can lead to inconsistencies. This often results in artifacts at the junctions of these models, affecting the realism and coherence of the generated poses.

To address the limitations of the SMPL-X model, a deep learning approach that comprehends, and represents each body part at a finer level is essential. Such a method should not only learn the nuances of individual body parts, but also integrate them effectively to capture the full spectrum of human body intercorrelations. In the pursuit of this goal, we will introduce PIXIE [12] in the subsequent section. PIXIE stands out for its ability to generate stable, and coherent 3D human models.

3.1.3 PIXIE: An Expressive Bodies

PIXIE is a model designed for reconstructing SMPL-X humans with realistic faces from a single RGB image. It utilizes specialized sub-networks for the body, face/head, and hands, integrating these components within a comprehensive network architecture for effective human form reconstruction.

PIXIE utilized the SMPL-X body model, adept at capturing the entire body’s pose and shape, including intricate facial expressions, and detailed finger articulation. This model is a differentiable function, denoted as $M(\beta, \theta, \psi)$, when parameterized by shape (β), pose (θ), and expression (ψ), outputs a detailed 3D mesh.

- **Shape Parameters** (β): The model employs 200 shape parameters that define the body, face, and hands. These parameters are coefficients in a linear shape space, learned from a set of registered body scans. This unified shape space naturally encapsulates the correlations among the body, face, and hand shapes.
- **Expression Parameters** (ψ): Facial expressions are modulated by 50 coefficients in a low-dimensional linear space, providing nuanced control over facial expressions.
- **Pose Parameters** (θ): The comprehensive pose parameters encompass the body, jaw, and hand poses. Each joint rotation, except for the jaw, is encoded as a 6D vector, ensuring a detailed representation of body posture. The jaw rotation, due to its unique movement range, is represented by a 3D vector using Euler angles.
- **Joint Notation**: The model articulates the posed joints as $X(\theta, \beta)$ in a structured format, accommodating a total of 55 joints, ensuring a detailed and comprehensive representation of the human form.

For image-based reconstruction, they adopted a weak-perspective camera model characterized by scale (s), and 2D translation (t). This model simplifies the projection of the 3D joints (X), and model vertices (M) onto the image plane, denoted as x and m respectively, facilitating the accurate transformation of 3D model data into a 2D image representation. PIXIE also incorporated implicit reasoning about gender to promote the generation of body shapes that align with typical gender characteristics. This feature is designed to enhance the realism and diversity of the generated human models. However, when applied to videos, the model may produce jittering or artifacts as it lacks training on motion or temporal sequences. This issue will be addressed in a subsequent method chapter, where we will introduce techniques as part of data preprocessing for smoothing and enhancing the temporal coherence of the generated 3D human model.

3.2 Cross-Modal Sequence Generation

Cross-Modal Sequence Generation is a significant area of research in the fields of natural language processing and computer vision, focusing on the conversion of data from one modality to another. The objective is to generate a sequence in one modality based on an input sequence in a different modality. This process involves understanding and mapping the relationships between different modalities to produce accurate and meaningful output sequences. Tasks such as text-to-speech (TTS) [78,

79, 80], speech-to-gesture [81, 82, 83], and image/video captioning (pixels-to-text) [84, 85, 86] all involve solving the Cross-Modal Sequence Generation problem. Initially, a combination of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) [87, 88, 88] were prevalent in addressing this problem. However, with the advent of the attention mechanism and its ability to capture long-range token interaction [89], transformer-based networks have started to outperform other sequence models, achieving top results in visual to audio [90, 91, 92], and audio to human motion [93, 94, 95] cross-modal sequence transformation generation tasks.

Two prominent cross-modal methods on which our work is built upon: the Full Attention Cross-Modal Transformer or FACT by Li et al. [10], proposed for generating realistic 3D dance motion from music, and Learning-to-Listen by Ng et al. [11], which focuses on developing a transformer-based model for generating listener facial response motion sequences. The Learning-to-Listen model is based on the architecture of the Full Attention Cross-Modal Transformer [10], capturing the long-range sequence correlation between the input data of the speaker’s facial motion and the corresponding audio, and the listener’s past facial motion to produce realistic 3D listener facial motion responses that are synchronous with the speaker.

3.2.1 FACT Transformer

The Full Attention Cross-Modal Transformer (FACT) by Li et al. [10] model is designed for the task of 3D dance motion generation, given a seed motion and audio features as inputs. The model architecture, shown in Figure 3-3, consists of three transformers: a motion transformer, an audio transformer, and a cross-modal transformer. The motion and audio inputs are first encoded using the motion transformer and audio transformer, respectively, into motion and audio embeddings. These embeddings are then concatenated and sent to the cross-modal transformer, which learns the correspondence between both modalities and generates N future motion sequences.

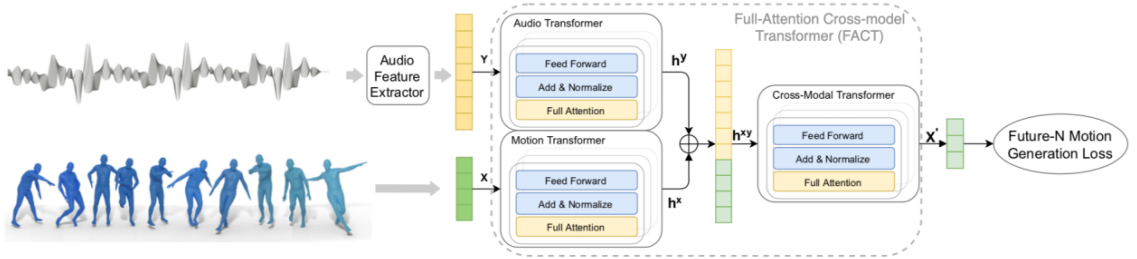


Figure 3-3: The Full Attention Cross-Modal Transformer (FACT) model architecture for generating 3D dance motion uses a seed motion of 2 second sequence and audio features as inputs, it autoregressively produces long-range future motion sequences that correspond to the input music. (source: Li et al. [10]).

FACT employs a full-attention mask in all three transformers [10, 96], allowing

internal tokens to have access to all inputs. This design choice makes the model more expressive than a causal model, and enables it to capture long-range dependencies between the motion and audio inputs. The full-attention mechanism also allows the model to be applied efficiently in an auto-regressive framework during testing. During training, it is only trained to predict the unseen future after the context window, rather than predicting the immediate next future token as in previous work [97, 94] that predict the immediate next future token. Specifically, the model is trained to predict N future motion sequences beyond the current input, which encourages the network to pay more attention to the temporal context. This design choice is critical in generating non-freezing, more realistic motion sequences.

FACT uses an early fusion strategy of fusing the motion and audio embeddings, followed by the use of a deep 12-layer cross-modal transformer module. This is in contrast to other works that used a single MLP to combine the embeddings [94]. The deep cross-modal module is essential for training a model that pays attention to the input music, which is particularly important in dance, where similar choreography can be used for multiple music. Without a deep cross-modal module, the network is prone to ignoring the conditioning music.

FACT is trained end-to-end in a self-supervised manner, using the generated N future motion sequences to train the model. At test time, the model is applied in an auto-regressive framework, where the first predicted motion is used as the input for the next generation step, and all conditioning is shifted by one.

3.2.2 Learning-To-Listen

The authors present a method for modeling conversational dynamics between a speaker and a listener by predicting the listener’s interactional facial motion responses. The goal is to capture the subtleties of face-to-face communication, and test whether the model can accurately represent the flow of conversation. The model takes multimodal inputs from the speaker, including 3D facial motion and audio, and uses cross-modal attention to fuse the two modalities using a similar cross-modal transformer of FACT, as described in the previous Section 3.2.1. The past listener facial motion is discretized using a novel sequence-encoding VQ-VAE, as illustrated in Figure 3-4. This approach enables the model to predict a distribution over the discrete codebook indices. The autoregressive predictor then samples a code for the next timestep and decodes it to obtain the continuous future listener facial motion. The overall model architecture, encompassing the VQ-VAE and the autoregressive predictor, is illustrated in Figure 3-5.

The task is defined as autoregressively predicting the future facial motion of the listener given the current 3D facial motion and audio of the speaker, along with the past listener motion. The model uses a transformer-based predictor, to learn correlation in temporally long-range patterns of the speaker and listener input sequences, a motion-audio cross-modal transformer to fuse the speaker’s audio and facial motion,

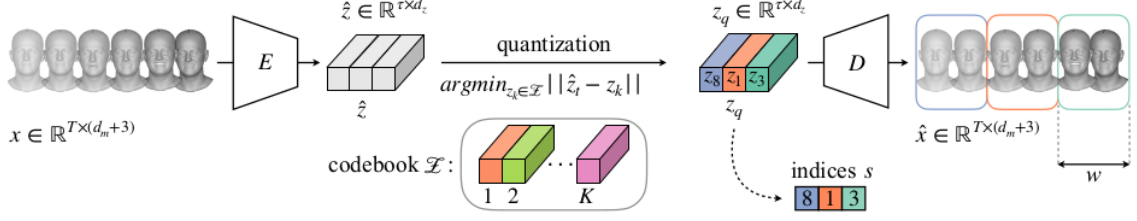


Figure 3-4: Discrete Listener Motion Codebook Learning using VQ-VAE: The input is a sequence of raw listener facial motion (expression coefficients and 3D head rotations) of length T . The transformer sequence-encoder E compresses the input into an embedding, which is then quantized and mapped to the closest codebook element in Z . The transformer decoder D reconstructs the input from the quantized embedding. (source: Ng et al. [11]).

and a VQ-VAE to represent the manifold of realistic listener facial motion. The output is a distribution over possible synchronous and realistic listener responses, from which multiple trajectories can be sampled.

The study employs 3D Morphable Face Models (3DMMs) [98, 99] to represent the 3D facial motion. This representation consists of three main components: facial expression parameters, shape coefficients, and head orientation represented using Euler angles.

To create an identity-agnostic representation, the model is trained without considering the shape coefficients. This step ensures that the output face shape remains consistent across all frames or sequences, allowing the model to generalize better for unseen listeners regardless of the individual’s unique facial features.

Furthermore, the head orientation is normalized to achieve a camera-view agnostic representation. This is accomplished by computing the average frontal face direction for each video, which essentially represents the rest pose for the individual in that video. Subsequently, all head poses in the sequence are aligned relative to this rest pose, ensuring consistency in head orientation across different video samples.

As a result, the 3D face representation becomes invariant to changes in facial shape, scale, and camera pose. This invariance enables the model to effectively generalize and adapt to new faces and camera viewpoints, enhancing its overall robustness and applicability.

3.3 Summary

In this chapter, we explored PIXIE as a versatile tool for 3D human modeling. Given a single image, PIXIE can accurately reconstruct 3D human mesh representations of body shape, pose, hand articulation, and facial expressions using SMPL-X parameters. This functionality enables us to create detailed representations of participants in the

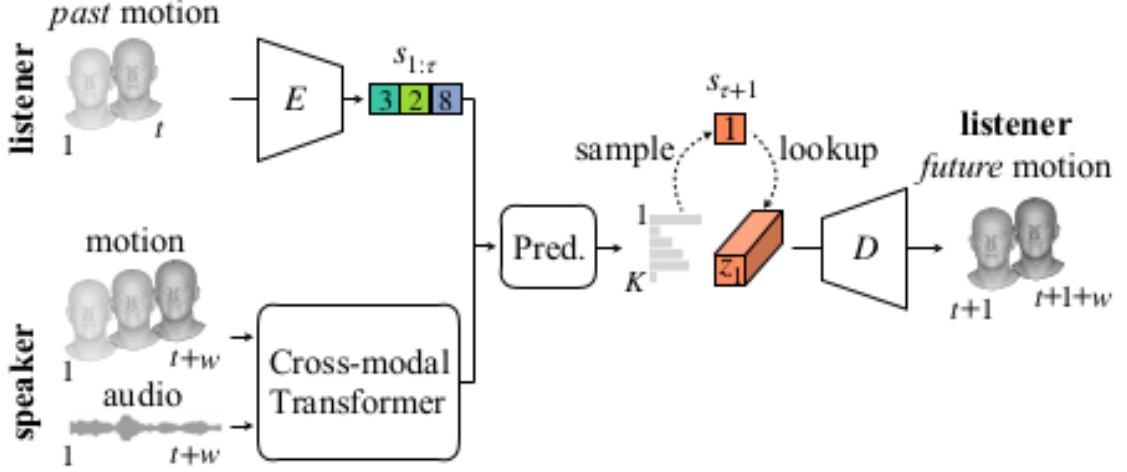


Figure 3-5: The presented method predicts a distribution of future listener motion based on multimodal inputs from a speaker. It employs cross-modal attention to combine the speaker’s audio and motion input and utilizes a novel sequence-encoding VQ-VAE to discretize the past listener motion. The autoregressive predictor (Pred.) generates a distribution over the discrete codebook indices, from which a code is sampled for the next timestep. Decoding the sampled codebook index yields the output future listener facial motion. (source: Ng et al. [11]).

images or videos, including nuanced body, hand movements, and facial expressions.

We introduced the Learning to Listen model, a transformer-based architecture built upon FACT, which focuses on predicting listener facial motion responses to the speaker in dyadic settings. By modeling the conversational dynamics between a speaker and a listener, this model offers valuable insights into the complex choreography of human interaction.

While the "Learning to Listen" model makes valuable contributions, it has limitations. Firstly, it focuses solely on facial expressions in dyadic interactions, neglecting the broader range of non-verbal cues present in real-world conversations. These cues also include body language and hand gestures, which significantly influence how a message is received. Secondly, the model doesn’t account for the role of personality traits in shaping listener responses. Research suggests that personality can influence the extent and type of engagement a listener displays [19, 20, 21]. For example, extroverts tend to exhibit more pronounced gestures, while introverts might have more reserved expressions and movements [citations] [24, 29, 30, 25, 31].

In order to address the existing limitations, the following chapter will delve deeper into the incorporation of a wider range of non-verbal cues in our approach. We plan to expand our model by including an extensive array of non-verbal behaviors such as facial expressions, body, and hand gestures. Furthermore, we aim to examine how a listener’s personality traits affect their non-verbal responses in dyadic conversations. By doing

so, we hope to capture the subtle complexities of one-on-one human communication more effectively, and refine our model to generate more realistic and engaging listener responses.

Chapter 4

Method

The previous chapter highlighted the foundation and limitations in existing listener response generation models, particularly their narrow focus on facial expressions in dyadic interactions. While the "Learning to Listen" model by Ng et al. [11] offered a valuable foundation for generating listener facial motion based on speaker cues, it overlooked the crucial role of other non-verbal communication channels. Real-world communication relies heavily on a richer tapestry of non-verbal signals, encompassing not just facial expressions, but also body language and hand gestures. Additionally, personality traits significantly influence how individuals engage in non-verbal communication.

This chapter presents our methodology for addressing these limitations. We build upon the transformer-based architecture of "Learning to Listen", and propose our approach for generating personality-aware listener in dyadic interactions. Specifically, we extend the model's scope beyond facial expressions to encompass a wider range of motion capture data, including facial expressions, upper body posture, arms, and hand gestures. Furthermore, we incorporate the listener's personality trait of extraversion into the model. This allows our model to generate listener responses that are not only reactive to the speaker's cues, but also reflect the listener's personality, fostering a more nuanced and realistic representation of listener behavior in dyadic interactions.

To clarify, by integrating upper body gestures and hand movements along with incorporating listener personality, we aim to enrich the model's output with non verbal behaviors that vary according to the listener's personality. This allows for a more differentiated depiction of listener responses, such as an extroverted listener employing larger gestures, and more direct engagement with the speaker, in contrast to an introverted listener who might exhibit subtler gestures and less direct engagement. These enhancements are expected to improve the model's depiction of dyadic interactions, offering more personalized and differentiated responses that better reflect individual listener characteristics.

This chapter is structured as follows: Section 4.1 details the conversion of raw data into a format suitable for our model, focusing on the preparatory steps required. Section 4.2 outlines the specific goal we address, by clarifying the inputs and outputs

of the model, and explains the notations employed. Finally, Section 4.3 explores the model workflow and architecture .

4.1 Data Preprocessing

The extraction of meaningful features for our model necessitates the thorough pre-processing of raw video data. As the first crucial step in our data pipeline, Figure 4-1 illustrates this process. Designed specifically for videos from dyadic recordings capturing interactions between two participants, the pipeline aims to transform these raw recordings into structured data features suitable for model input. The final data set includes sequences of motion features for both the listener and speaker, along with encoded audio sequences from the speaker.

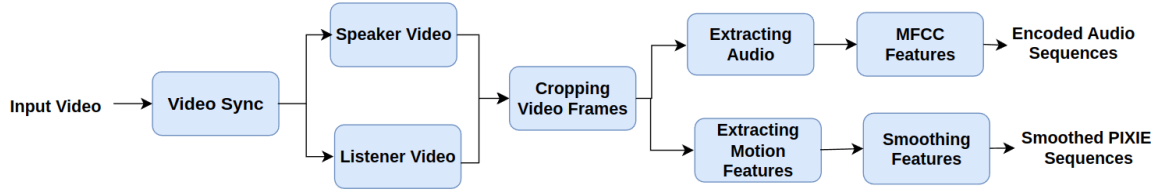


Figure 4-1: Our proposed pipeline for processing raw dyadic interaction videos. The pipeline processes raw dyadic interaction videos, segmenting them into separate speaker and listener video streams. For each video stream, motion sequence features are extracted, including PIXIE [12] parameters for facial expressions and upper body joints. Additionally, the speaker’s audio is encoded into MFCC sequence features. The final output is a comprehensive dataset containing speaker and listener motion features alongside speaker audio features, suitable for training our model.

The pipeline initiates by segmenting videos from dyadic interactions, involving two participants each, into separate streams for speakers and listeners. This segmentation utilizes timestamps from conversation transcripts, resulting in isolated speaker and listener videos for each participant.

To concentrate on the participants, we define a maximum boundary box around each using dataset annotations that mark body, face, and hand joint landmarks. The boundary is determined by the extreme points: the shoulders (leftmost and rightmost), the head (topmost), and the torso (bottommost). This step is vital for accurately cropping the video frames around the participants.

The audio sequence from the speaker’s video is extracted and processed using Librosa, converting it into a mel-spectrogram with a sampling rate of 16 kHz. For motion feature extraction, we employ PIXIE to process both speaker and listener videos. This process results in the extraction of SMPL-X parameters for each frame of the video sequence. These parameters encompass 50 blended shape coefficients for facial expressions, 3D jaw and head rotation, and 306 upper body parameters.

These parameters represent each body part, specifically focusing on the relative joint positions, all represented in Euler angles. To maintain consistency throughout the frame sequence and to ensure a gender-neutral model representation, we exclude the 200 body shape parameters.

During the rendering of motion features extracted by PIXIE, we observed jittering in the generated motion sequences. To address this issue, we applied a dual filtering strategy directly on the extracted parameters for each frame of the video sequence. Initially, we implemented a mean average filter within a window size of 7 frames. This filter calculated the mean average of the PIXIE parameters, which include 50 blended shape coefficients for facial expressions, 3D jaw and head rotation, and 306 upper body parameters representing relative joint positions in Euler angles. The mean average was then used to replace the current frame’s parameters, ensuring a smoother transition between frames while preserving the overall motion characteristics. Subsequently, to minimize abrupt motion jumps while retaining motion variability, we employed a median filter with a smaller window size of 3 frames, which was applied to the same set of PIXIE parameters. This careful selection of window sizes for each filter allowed us to effectively mitigate jittering without sacrificing the richness of motion detail, resulting in smoother and more natural motion sequences. The final output is a refined dataset, consisting of speaker audio features and motion features for both speaker and listener, ready for model training.

4.2 Problem Formulation

For each timestep $t \in [1, T]$, given the past listener frames sequence of facial motions $\hat{F}_{1:t-1}^L$ and upper body motions $\hat{B}_{1:t-1}^L$, along with the speaker’s sequence of facial motions $F_{1:t}^S$ and upper body motions $B_{1:t}^S$. We also consider the speaker’s audio sequence $A_{1:t}^S$. Our objective is to generate a listener’s future motion sequence, conditioned on the corresponding input listener’s extraversion trait score. The future sequence is generated autoregressively, one time-step at a time, as both facial and upper body motion sequences, denoted by $\hat{f}_t^L$ and $\hat{b}_t^L$, respectively.

4.3 Model Architecture

In this section, we present our personality-aware model architecture, illustrated in Figure 4-2. The model employs a transformer-based predictor $\mathcal{P}$ (Section 4.3.4), which autoregressively generates the listener’s future motion sequence for both facial and upper body, denoted by $\hat{f}_t^L$ and $\hat{b}_t^L$, at each timestep t . The predictor captures the correlation between the temporal sequence patterns of both the speaker and listener.

The speaker’s input consists of sequences of facial $F_{1:t}^S$, upper body motion features $B_{1:t}^S$, and audio $A_{1:t}^S$. The motion features are extracted using PIXIE from the corresponding speaker video segments as described in Section 4.1, which includes a set

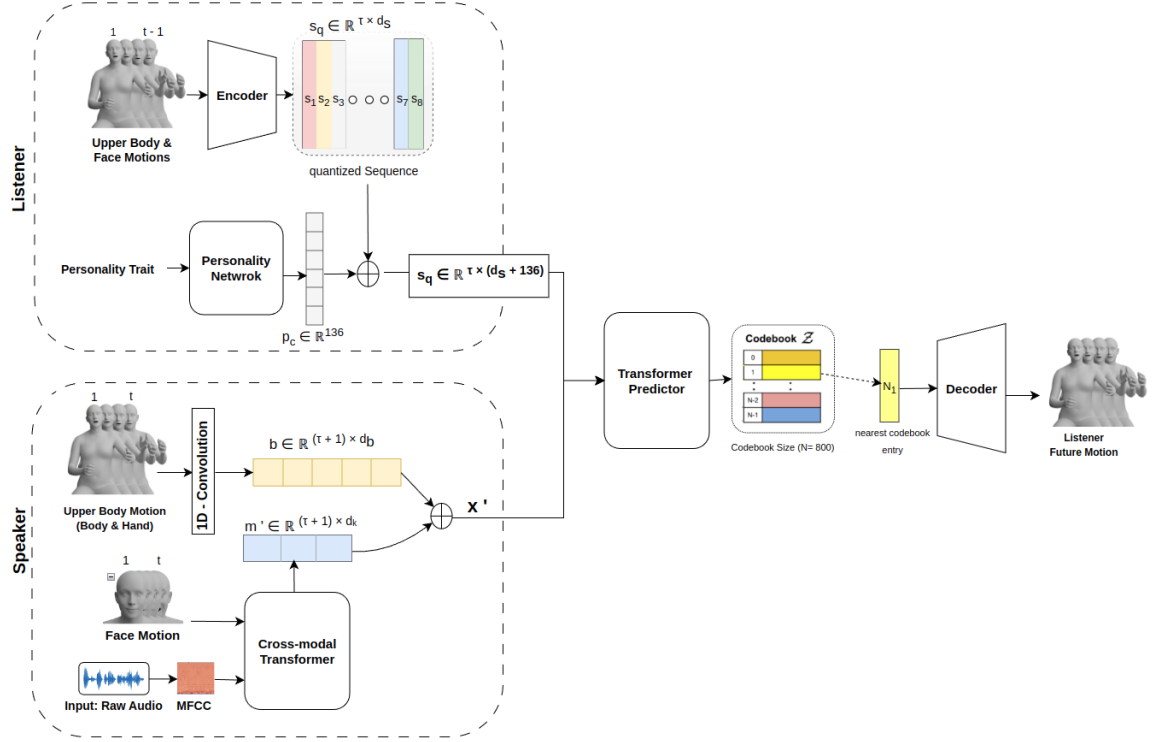


Figure 4-2: Model Architecture: Our method generates a listener response, conditioned on multimodal speaker inputs and individual listener personality traits. The model employs a transformer-based predictor that captures the correlation between the temporal sequence patterns of both the speaker and listener. The autoregressive predictor forecasts future listener motion, creating a nuanced, personality-aware response.

of parameters such as 50 blended shape coefficients for facial expressions, 3D jaw and head rotation, and 306 upper body parameters. The audio is represented as a sequence of mel-spectrogram frequencies. The facial motion and audio sequences are integrated using a cross-modal transformer (Section 4.3.3) to capture long-range dependencies between the motion and audio inputs. The fused output is a temporally aligned fusion of audio and facial motion sequences of length $\tau = 8$. This output is concatenated with the corresponding upper body motion features to form the final speaker input for the predictor transformer $\mathcal{P}$ (Section 4.3.4). The 1-D convolution layer in the model, shown in Figure 4-2, is used to downsample the upper body input sequences to match the cross-modal transformer output length τ .

The listener’s input consists of both facial motions $\hat{F}_{1:t-1}^L$ and upper body motions $\hat{B}_{1:t-1}^L$ sequences, along with the corresponding listener’s personality/extraversion score. This score is a scalar value representing the different levels of extraversion for the listener. The personality score is encoded using a Personality Network (Section 4.3.2), which produces a conditioned vector p_c . This vector is concatenated with the feature dimension space of the corresponding listener’s quantized sequence s_q , thereby embedding personality traits directly into the model’s input. The quantized sequence s_q is a compact representation of the listener’s facial and upper body motions features within the learned discrete latent space of the codebook (Section 4.3.1). The listener’s quantized sequence s_q concatenated with the corresponding personality vector p_c is passed along with the speaker’s input via the predictor transformer $\mathcal{P}$ to predict the listener’s future sequence. The predictor’s output is matched against the closest codebook entry, and the final quantized sequence is decoded to represent the final listener output sequence.

4.3.1 Quantized Listener Motion Codebooks

To capture the rich dynamics of non-verbal communication cues, we utilize a vector quantized variational autoencoder (VQ-VAE) model, inspired by [11]. However, we extend their approach by incorporating both facial and upper body motions, extracted from listener video frames within our dataset using PIXIE [12]. This expansion allows for a more comprehensive representation of non-verbal cues, encompassing subtle facial expressions, intricate body language, and nuanced gestures. This model’s architecture, depicted in Figure 4-3, is well-suited for temporal sequences like listener motions due to its utilization of transformers for both the encoder and decoder. This enables the model to effectively learn long-range dependencies and capture high-frequency movements.

The VQ-VAE model’s core operation involves three main steps: encoding, quantization, and reconstruction.

1. **Encoding:** downsamples the input listener motion sequence using convolutional layers to reduce computational complexity while retaining essential features. The transformer architecture then enriches this compressed representation with

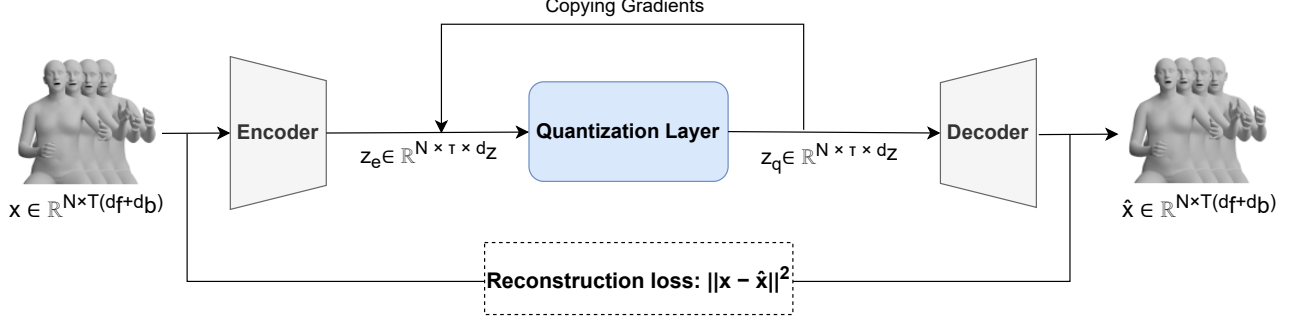


Figure 4-3: Listener Motion VQ-VAE: A Model for Quantizing Listener’s Facial Expressions and Upper Body Motions into a Discrete Codebook.

temporal dynamics and motion relationships through positional embeddings. This process enables the model to efficiently capture complex listener motion patterns and learn long-range dependencies and high-frequency movements in the input sequence.

2. **Quantization:** the continuous latent representation learned by the encoder is quantized to a discrete code using a pre-defined codebook, which acts like a dictionary of common motion patterns observed in the training data. During quantization, the model finds the code in the codebook that best represents the continuous latent representation, essentially ”snapping” it to the nearest discrete code. This focus on discrete codes offers several advantages, particularly addressing the challenge of drifting often encountered with continuous representations. Drifting occurs when the model’s predictions gradually deviate from realistic motion patterns during training. By selecting from a pre-defined set of codes that represent valid motion patterns, VQ-VAEs are less susceptible to this issue. The codebook acts as a guiding force, encouraging the model to explore a wider range of meaningful motion patterns, and preventing it from venturing into unrealistic regions of the latent space.
3. **Reconstruction:** the decoder reconstructs the motion sequence, generating listener motions that closely resemble the original.

The model’s pipeline, illustrated in Figure 4-3, begins by encoding a batch of N listener sequences. Each sequence comprises facial motions $F_{1:T}^L \in \mathbb{R}^{T \times d_f}$ and upper body motions $B_{1:T}^L \in \mathbb{R}^{T \times d_b}$, collectively represented as x . Here, T denotes the length of the timestamp sequence, while d_f and d_b represent the dimensions of the facial and upper body features, respectively.

The encoder consists of two convolutional layers, each succeeded by max pooling.

Given an input listener sequence x with a length of $T = 32$, the sequence is reduced to a length of $\tau = 8$ after the convolutional stages. This condensed sequence is then processed by a transformer, which comprises 12 hidden layers with a hidden size of 500 and 10 attention heads. This architecture allows the encoder to condense longer motion sequences into shorter segments, aiding in the recognition of long-range dependencies. It shifts the focus from individual face expressions and body movements to sequences, enabling the latent embedding to cover multiple frames. This approach helps in understanding the motion dynamics more effectively. The output of the encoded sequences is denoted as:

$$z_e = E(x) \in \mathbb{R}^{N \times \tau \times d_z}$$

Where N is the number of sequences, $\tau = 8$ (is the length of the patch-wise encoded sequence), and d_z is the output features dimension, resulting from the linear embedding of input facial features d_f and body features d_b into a larger space to capture more contextual information.

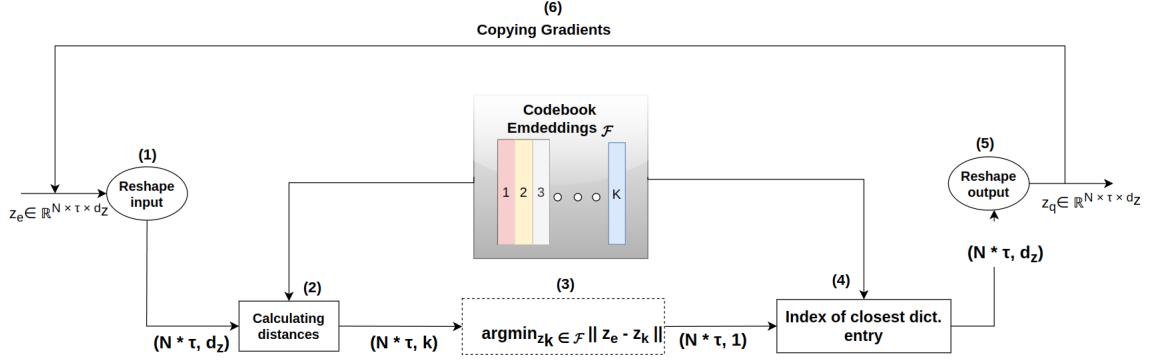


Figure 4-4: Quantization Layer: Illustrating the Steps of the VQ-VAE Quantization Process. we adopted the Learning-to-Listen VQ-VAE architecture [11]. We employed a codebook of size 800 and quantized embedding vectors with a dimension of 500. To achieve a more manageable input sequence, we downsampled it using 2 convolution layers, resulting in a quantized sequence of length 8, which is twice the length used in the original Learning-to-Listen model. Additionally, we utilized an encoder transformer with 10 attention heads to effectively capture the correlation patterns in the face and upper body features sequences.

Proceeding with the quantization process, the Quantization Layer transforms the encoded sequences z_e into a quantized sequences z_q , following a series of steps as illustrated in Figure 4-4:

1. **Reshape input**: all dimensions, except the final feature dimension d_z , are merged into a single dimension, resulting in $N \times \tau$ vectors, each with a dimensionality of d_z .

2. **Calculating distances:** for each of the $N \times \tau$ vectors, we compute the distance to each of the k vectors in the codebook dictionary $\mathcal{F}$, yielding a distance matrix with dimensions $(N \times \tau, k)$.
3. **Quantization function:** for each of the $N \times \tau$ vectors we find the index of closest of the k vectors from dictionary $\mathcal{F}$ via an element-wise quantization function:
$$z_q = q(z_e) := \arg \min_{z_k \in \mathcal{F}} \|z_e - z_k\|, \quad z_q \in \mathbb{R}^{N \times \tau \times d_z}$$
4. **Index of closest embedding:** index the closest vector from the dictionary for each of $N \times \tau$ vectors.
5. **Reshape Output:** Convert the output back to the original input shape (N, τ, d_z) , resulting in the final quantized sequences $z_q \in \mathbb{R}^{N \times \tau \times d_z}$.
6. **Copying gradients:** during training, the gradient from the decoder is copied back to the encoder’s output, bypassing the non-differentiable argmin operation in the quantization step, thus enabling effective backpropagation and learning.

Finally, the quantized sequences z_q pass through the decoder, which mirrors the encoder’s architecture. It employs transposed convolutions to reconstruct the listener motion sequences, approximating the original input ($\hat{x} \approx x$):

$$\hat{x} = D(z_q) \in \mathbb{R}^{N \times T(d_f + d_b)}$$

4.3.1.1 Loss Function

The Encoder E , the Decoder D , and the codebook $\mathcal{F}$ are trained using the following composite loss function:

$$\begin{aligned} L_{VQ}(E, D, \mathcal{F}) = & \|x - \hat{x}\|^2 && \text{(reconstruction loss)} \\ & + \|\text{sg}[E(x)] - z_q\|^2 && \text{(codebook loss)} \\ & + \|\text{sg}[z_q] - E(x)\|^2 && \text{(commitment loss)} \end{aligned} \quad (4.1)$$

Where:

- The **reconstruction loss** quantifies the fidelity of the reconstructed data $\hat{x}$ to the original input x by measuring the L2 distance for each set of features, including 50 facial expression parameters, 3D head orientation, 3D jaw orientation, and 306 upper body parameters (encompassing left wrist joints, right wrist joints, left hand joints, and right hand joints). We applied this loss separately to each set of features and then combined them to get the total reconstruction loss. This approach allowed the model to learn more fine-grained details and produce more accurate reconstructions of the facial expressions and body movements, ensuring that the output remained a true representation of the input.

- The **codebook loss**: This loss ensures that the codebook embeddings are useful and representative of the encoder outputs. Specifically, it encourages the embeddings e_i within the codebook to move closer to the encoder output $E(x)$.
- The **commitment loss**: This loss serves as a regularizer to encourage the encoder to produce outputs that are close to the codebook embeddings. Specifically, it penalizes the encoder if the chosen embedding vector e_i for an input x strays too far or drifted from the encoder output $E(x)$.

4.3.2 Listener Personality Network

In order to incorporate listener personality, we utilize a multi-layer perceptron (MLP) network. The network takes as input the personality scores of extraversion corresponding to each listener sequence and generates a conditioned vector p_c with a length of 136. Initially, we tried to directly concatenate the personality values without embedding them to a vector, but we found that the current approach yielded better results. It is important to note that we train this network with the full model and not independently. This ensures that the network learns to generate conditioned vectors that are compatible with the rest of the model’s architecture.

4.3.3 Cross-Modal Attention for Speaker Input

In processing the speaker’s input, we utilize facial motion data $m = F_{1:t}^S$, and the corresponding audio $a = A_{1:t}^S$. To effectively merge these two modalities, we apply a cross-modal attention transformer, with same architecture by Ng et al. [11]. who advanced upon existing cross-modal attention frameworks. Specifically, Ng et al. [11] extended the application of cross-modal attention, previously focused on text-audio [100] and language-vision [101, 102], to include the fusion of audio and motion modalities. Unlike some previous methods, such as [35] that processed motion and audio separately using distinct encoders, our approach specifically focuses on integrating motion and audio modalities based on [11]. Additionally, applying cross-modal attention within temporal sequencing not only integrates different modalities, but also facilitates the exploration of potential temporal realignments between them [100].

In the model in Figure 4-5, speaker inputs of different modalities are first processed through individual linear layers to create a uniform feature’s representation. These processed inputs are then fed into the cross-modal transformer. Within this transformer, a stack of cross-attention blocks operates, with audio serving as the ‘Queries’ Q_a to sift through the motion ‘Keys’ K_m and extract pertinent ‘Values’ V_m . The cross-modal attention mechanism applies a softmax function to the dot product of the queries and keys, scaled by the square root of the transformer hidden size d_k , and uses these scores to select and weight the motion values:

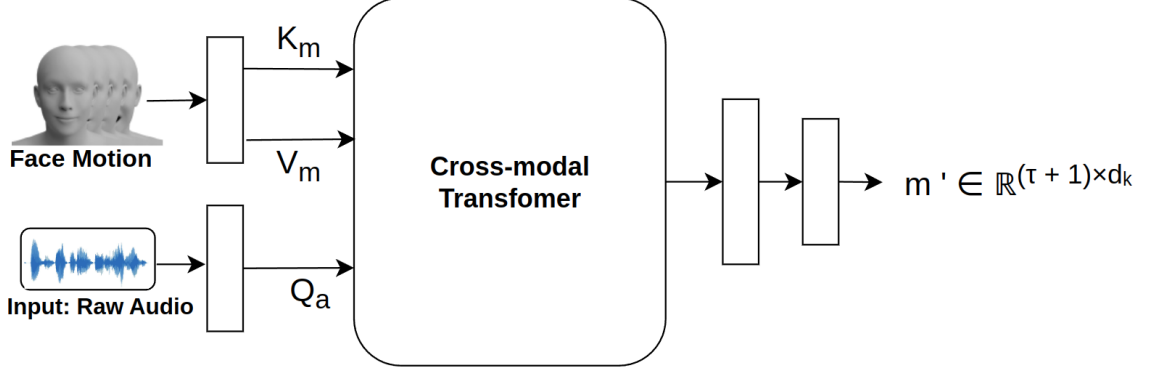


Figure 4-5: Cross-modal Transformer: Speaker inputs are integrated in a cross-modal transformer, where each modality is initially processed through a linear layer. Audio forms queries, and motion data act as keys and values in cross-attention blocks, followed by 1D convolutions to align the output’s sequence with the quantized listener input.

$$\text{attention}_{m \rightarrow a} = \text{softmax} \left(\frac{Q_a K_m^T}{\sqrt{d_k}} \right) V_m \quad (4.2)$$

The outcome of this attention process is a temporally aligned fusion of audio and facial motion features. To finalize the integration, 1D convolutions are applied to downsample the temporally fused output, ensuring it aligns with the quantized listener input’s temporal sequence $\tau = 8$. The processed output $m' \in \mathbb{R}^{(\tau+1) \times d_k}$, which represents an integrated fusion of facial motion data and audio information, is then concatenated with the corresponding features of the speaker’s upper body sequence motion $b \in \mathbb{R}^{(\tau+1) \times d_b}$, which is also down-sampled using a 1D convolution to match with m' . This combined representation serves as the final speaker input $x' \in \mathbb{R}^{(\tau+1) \times (d_k + d_b)}$ to the listener predictor module in next Section.

4.3.4 Listener Motion Predictor

Up to this point, we have prepared our input sequences for both listeners and speakers. These sequences will serve as inputs to the Transformer Predictor module depicted in Figure 4.3. More precisely, the Predictor module $\mathcal{P}$, receives the final speaker input x' . Additionally, it takes in the quantized past motion sequence from the listener, concatenated with conditioned output vector of the personality network (as explained in Section 4.3.2). The Listener input $s_q \in \mathbb{R}^{\tau \times (d_s + 136)}$, conditioned on the personality, significantly contributes to the final model’s generation as the personality embeddings incorporated additional information into the listener’s motion data as we will see in the final results of our model evaluations (see Section 5.5, and Section 5.3 of the evaluation chapter).

We represent the past listener’s quantized motion of length $\tau = 8$ as $s = s_{1:\tau} \in$

$\{1, \dots, K\}^\tau$, where K is the codebook size. After training the code dictionary using VQ-VAE (as described in Section 4.3.1), we utilize the pre-trained encoder E and quantization q to discretize the past input listener’s face $\hat{F}_{L1:t-1}$, and upper body motions $\hat{B}_{L1:t-1}$). This process returns a sequence of indices representing the nearest codebook entry for each element. You can visualize this in Figure 4.3 of the model as the quantized or discretize sequence $s = s_{1:\tau}$ block following the listener encoder.

To convert the given discretized sequence of indices $s_{1:\tau} \in \mathbb{R}^{1 \times \tau}$ into embeddings $s_q \in \mathbb{R}^{\tau \times d_s}$, we utilize the token or dictionary indices of the sequence, and apply an embedding function using PyTorch. Additionally, we incorporate position embeddings, since transformers inherently lack positional information, which assists in capturing the relative positions of codebook indices within the sequence. We concatenate each listener embedded sequence $s_q \in \mathbb{R}^{\tau \times d_s}$ features with its corresponding personality conditional vector $p_c^L \in \mathbb{R}^{136}$ that result to a unified listener input for predictor $\mathcal{P}$ incorporating each listener personality encodings.

To convert the given discretized sequence of indices $s_{1:\tau} \in \mathbb{R}^{1 \times \tau}$ into embeddings $s_q \in \mathbb{R}^{\tau \times d_s}$, we utilize the token or dictionary indices of the sequence, and apply an embedding function using PyTorch. Additionally, we incorporate position embeddings since transformers inherently lack positional information, which assists in capturing the relative positions of codebook indices within the sequence. We concatenate each listener embedded sequence $s_q \in \mathbb{R}^{\tau \times d_s}$ feature with its corresponding personality conditional vector $p_{cL} \in \mathbb{R}^{136}$ feature. This results in a unified listener input $s_q \in \mathbb{R}^{\tau \times (d_s + 136)}$ for the predictor P , by incorporating each listener personality encoding. Given both listener $s_q \in \mathbb{R}^{\tau \times (d_s + 136)}$, and the speaker x' . We then concatenated both inputs sequences to get a sequence $s_q \in \mathbb{R}^{2 \times \tau \times d_s}$. This combined sequence serves as the input for the predictor. Since the Transformer is a set-to-set operation, it outputs a sequence of length 2τ matching the input. During training, we take the first 8 indices of the output, which correspond to the listener, and discard the remainder.

The multinomial distribution of the next listener codebook index across the K entries is denoted as $p(s_{\tau+1})$, and is calculated based on the speaker sequence x' and the listener sequence $s_q \in \mathbb{R}^{\tau \times (d_s + 136)}$:

$$p(s_{\tau+1}) = \mathcal{P}(x', s_q \in \mathbb{R}^{\tau \times (d_s + 136)}) \quad (4.3)$$

This distribution represents the likelihood of the next listener codebook index given the provided sequences. Subsequently, we utilize the decoder to generate a continuous prediction for the listener’s future motion. It’s essential to note that during this decoding process, we exclusively employ only the initial d_s features of the sequence, which specifically capture the predicted listener motion. The remaining 136 features contain personality-related information obtained during training, only for guiding the motion generation.

We train our predictor model in Figure 4.3 with a cross-entropy loss on the

codebook index $s_{\tau+1}$:

$$L_P = - \sum_y p(y) \log(p(s_{\tau+1})) \quad (4.4)$$

Where:

- L_P represents the cross-entropy loss.
- y represents the ground truth future facial motion and upper body motion.
- $p(y)$ represents the true probability distribution over y .
- $p(s_{\tau+1})$ represents the predicted probability distribution over the codebook index $s_{\tau+1}$.

During the training phase, we employ a technique known as teacher-forcing, where we provide the actual listener motion data, denoted as "y," as the past listener input. To promote autoregressive learning, we introduce random masking for previous time steps within the range of $[1, \tau]$. However, during the testing phase, the process differs.

We input zero values for the time steps that lack prior listener predictions and adjust the masking accordingly to exclude these steps. This approach enables us to predict future listener motion autoregressively for inputs of varying lengths. Importantly, during testing, the network does not have access to any ground truth past listener motion data.

Chapter 5

Experimental Evaluation

In this chapter, we introduce the dataset used for training and evaluating our models, followed by the experiment setup and model implementation details. We then introduce the evaluation metrics used to assess our model’s performance. The Quantitative Results section presents our model’s performance compared to baseline models, demonstrating the effectiveness of our approach in predicting listener behavior based on both the speaker’s behavior and listener’s personality traits. Finally, we describe two user studies conducted to evaluate our model’s performance from a human perspective.

5.1 UDIVA Dataset

UDIVA is a multimodal dataset [13] of face-to-face dyadic interactions. It has been collected using multiple audio-visual and physiological sensors, and includes contextual metadata (e.g., gender, age, and education level) as well as self-reported personality (OCEAN) traits. To the best of our knowledge, there is no similar publicly available face-to-face dyadic dataset in the research field in terms of the number of participants, recorded sessions, and contextual or personality data.

Currently, we have only access to UDIVA v0.5, which is a preliminary version of the UDIVA dataset. It includes a subset of participants, sessions, synchronized views, and annotations from the complete UDIVA dataset. UDIVA v0.5 consists of 145 dyadic interaction sessions, categorized into four different tasks (Figure 5-1): Talk, Lego, Ghost, and Animals. These sessions involve 134 participants, with ages ranging from 17 to 75 years (55.2% male). Each participant can engage in up to five sessions with different partners. The majority of interactions are in Spanish (73.1%), followed by Catalan (17.25%) and English (9.65%). [13]

For our study, we focus exclusively on dyadic interactions in the talk task, and we do not consider data from other tasks in our experiments. In the talk task, participants were instructed to engage in a 5-minute conversation on any subject. This task allowed for the analysis of common conversation constructs, such as turn-taking, synchrony, empathy, and quality of interaction, among others. During the recorded dyadic sessions



Figure 5-1: Overview of different UDIVA v0.5 tasks from 5 video sessions. From left to right: **Talk**, Lego, Animals, Ghost, Gaze. We only trained our model on dyadic sessions from the Talk task. [13]

of the talk task, two participants sit facing each other at a 90° angle around a table, as depicted in Figure 5-1, while engaging in a conversation. Those recordings are captured from different viewpoints using multiple cameras, as seen in Figure 5-2.

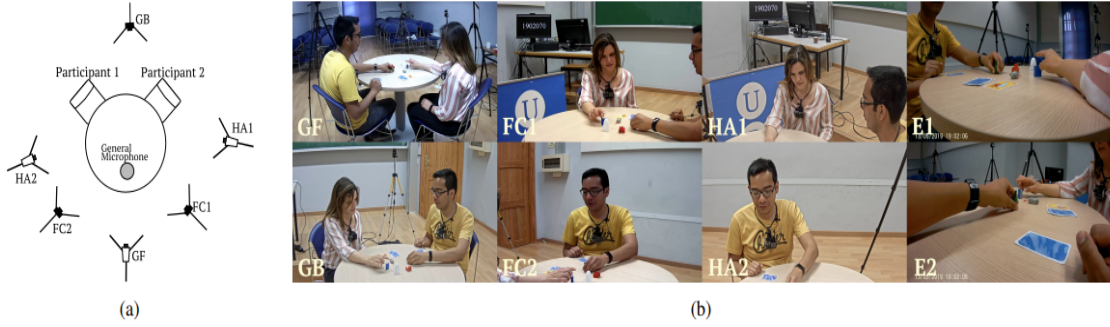


Figure 5-2: Multiple camera angles capture the same dyadic session featuring two participants. In UDIVA v0.5, only the frontal camera views, **FC1** and **FC2**, are available, each focusing exclusively on one of the two participants in every individual video.[13]

The dataset includes conversation transcripts, where participants are referred to as 'Participant 1' (visible in FC1 view) and 'Participant 2' (visible in FC2 view). These transcripts, accompanied by timestamps, aid in extracting video segments from both FC1 and FC2 during data preprocessing in Section 4.1. These segments correspond to moments when each participant is either speaking or listening.

Additionally, the dataset provides automatically extracted annotations for both participants in every session (Figure 5-3). These annotations include hand landmarks (21 landmarks, 3 coordinates), body landmarks (24 landmarks, 3 coordinates), facial landmarks (68 landmarks, 3 coordinates), and 3D eye gaze vectors (an array of 3 floats representing the 3D gaze vector). We leverage these annotations to determine maximum boundary boxes for cropping video frames (as described in Section 4.1) from both FC1 and FC2 videos around each participant, ensuring minimal interference.

The dataset also incorporates contextual metadata and personality information.

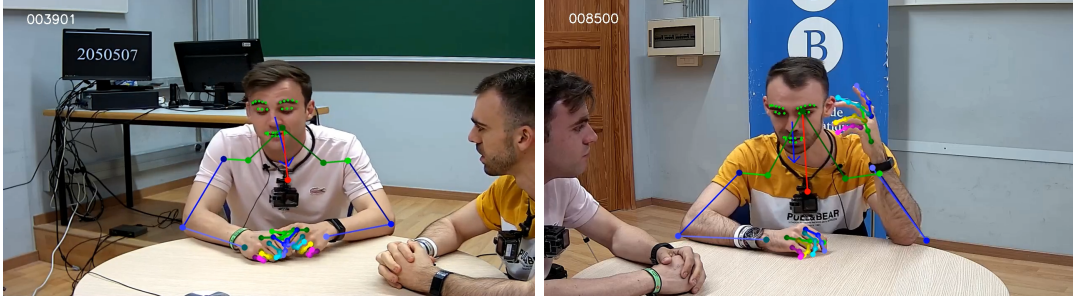


Figure 5-3: UDIVA Annotations Extracted from Videos of Two Participants (FC1 and FC2) in the Same Dyadic Session. Annotations files are available for reference.

Each participant in the dataset is assigned a unique identifier (ID), and their corresponding attributes, such as gender, age, and education level, are recorded alongside this identifier. Self-reported personality traits based on the OCEAN model 2.1.2 are also collected through surveys or questionnaires and linked to the respective participants' identifiers. We utilized the personality traits, specifically focusing on extraversion, associated with each participant as input for our models. Figure 5-4 illustrates the distribution of extraversion scores in our train set.

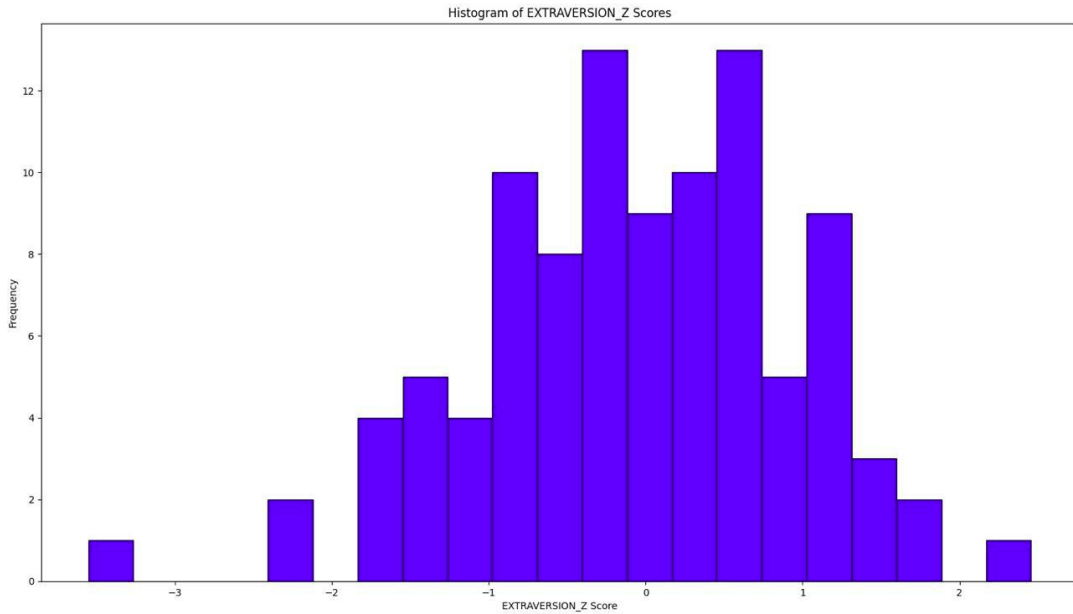


Figure 5-4: Distribution of Extraversion Scores: The figure shows the distribution of extraversion scores in the train set, where the x-axis represents the score values and the y-axis indicates the number of participants. it is important to note that this graph is generated using data from the UDIVA dataset.

To prepare the data for our model, we reorganized it based on participant identifiers instead of sessions. Our goal was to consistently position each listener participant on the left side in all their sessions, with their corresponding different speaking partners

on the right. When necessary, we mirrored videos to maintain these positions. This process resulted in participant-specific folders, each containing four videos: listener and speaker videos for the participant on the left (e.g., p0_listening and p0_speaking), and their corresponding session partner’s videos on the right (e.g., p1_listening and p1_speaking).

Following this arrangement, we applied the data preprocessing procedures outlined in Section 4.1 to these videos. This allowed us to extract listener and speaker motion sequences, as well as the corresponding encoded audio sequences of the speaker using MFCC frequencies [103]. The listener and speaker motion sequences consisted of PIXIE [12] face parameters, including 50 facial expression coefficients and 3D jaw Euler angles, along with PIXIE [12] upper body 306 parameters that are representing relative body and hand joints, and overall body pose. Additionally, we encountered 200 body shape PIXIE coefficients [12], which were discarded to create an identity-agnostic representation with a neutral gender. To maintain consistency across the dataset, we normalized both face and upper body parameters by calculating their mean and standard deviation before commencing the training process.

5.2 Experiments

We conducted all our experiments using the UDIVA v0.5 dataset [13]. The dataset is divided into three sets: training (99 participants), validation (21 participants), and testing (15 participants). During training, our model learns from a diverse group of listeners, encompassing all 99 participants in the training set. Specifically, we utilize data extracted from each listener participant (p0_listening) along with corresponding speaking partner (p1_speaking). For validation purposes, we employ the corresponding validation set to assess our model’s performance. In the testing phase, we evaluate our model on unseen listeners from the test set using both a set of quantitative metrics 5.3 and a perceptual user study 5.4.2.

5.2.1 Implementation Details

Our model in Figure 4-2 training consisted of two key stages. In the initial stage, we trained the VQ-VAE model to learn the listener codebook, as elaborated in 4.3.1. . Subsequently, in the second stage, we trained our predictor (detailed in Section 4.3.4) to generate future listener motions sequences. This involved quantizing past listener motion sequences, incorporating personality network input from Section 4.3.2, and integrating corresponding speaker-derived motion and audio features using the cross-transformer explained in Section 4.3.3. We will discuss some implementation details of our model’s training as follows, with reference to Figure 4-2:

- A. VQ-VAE: (learning the listener codebook)** We optimize our VQ-VAE model using AdamW optimizer [104, 105], employing a scheduled learning rate

decay with an initial rate of 0.1, a warm-up phase of 100,000 steps, and a batch size of 32. Our training dataset consisted of listener sequences of length 32. We trained the model for 2000 epochs, lasting around 3 days on 4 GPUs.

We employed a codebook $\mathcal{F}$ of size 800, and quantized embedding vectors with a dimension of 500. The encoder was constructed using 2 convolutional layers, each with a kernel size of 5, stride 1, and padding 2, followed by a max-pooling layer to downsample the input listener sequences, which initially had a length of 32, resulting in a sequence length of 8. Subsequently, we performed a linear embedding step to project the features into a higher-dimensional space, allowing for a more comprehensive understanding of the contextual information within our data. To preserve positional information after downsampling, we incorporated positional embeddings. Finally, the sequences were passed through a Transformer architecture, featuring a hidden size of 500, 10 attention heads, and 12 layers. The encoder, quantizer, and decoder components of the VQ-VAE model were trained jointly, optimizing the loss function outlined in Section 4.3.1.1.

B. Personality Network: (incorporating listener personality) We implement a multi-layer perceptron (MLP) network. This MLP network comprises three layers: an input layer, a hidden layer with 68 neurons, and an output layer with 136 neurons. The GELU activation function is employed within this network.

The network’s output is a conditional feature vector with a dimensionality of 136, which is concatenated with the corresponding input quantized listener sequence’s motion features. This concatenated vector serves as a conditioning factor for predicting the future listener’s motion based on personality traits.

Throughout our experiments, we explored various sizes for the conditional features, and determined that a vector size of 136 yielded the best results, aligning well with our model’s requirements.

C. Cross-modal Transformer: (modeling the speaker input) We utilize this transformer to integrate both speaker facial motion sequences, and their corresponding audio inputs. While the speaker’s motion sequence matches the input listener’s of length of 32, the audio sequence spans 128 time steps due to audio processing using the librosa library. Operating at a 16000 Hz sampling rate and generating mel spectrograms, the audio sequence is four times longer than the motion sequence. To address this discrepancy, we employ max pooling to downsample the audio, aligning its sequence length with that of the motion data. Subsequently, each of the facial motion and audio sequences undergoes projection to an expanded feature space through an embedding linear layer. Furthermore, positional embedding is applied to both sequences. Finally, these two sequences are concatenated and fed into the cross-modal transformer, as depicted in Figure 4-5.

The cross-modal transformer is a transformer architecture with a hidden size of 1024, featuring 10 attention heads and 12 layers. Following the transformer, there are two convolutional layers with a kernel size of 5, a stride of 1, and padding of 2. These convolutional layers are succeeded by max-pooling operations, which effectively shorten the length of the speaker embeddings from 32 to align with the corresponding quantized listener sequence, which has a length of 8. This alignment ensures temporal synchronization between both listener and speaker sequences.

To integrate the output from the cross-transformer, which now has a length of 8, with the speaker’s upper body motion sequences, we follow a similar process. We apply convolutional layers to downsample the upper body sequences to match the transformer output length, and then we utilize linear and positional embeddings. Finally, we concatenate these embeddings to create a comprehensive speaker input for the predictor.

- D. Predictor: (predicting future listener sequences)** We trained the predictor jointly with the cross-modal transformer for 1200 epochs, which took nearly 1 day on 4 GPUs. We used a learning rate of 0.001 with 50,000 warm-up steps and a batch size of 32.

The predictor is a full attention transformer with a hidden size of 938, 12 attention heads, and 5 layers. We concatenate the embeddings from both input speaker and listener sequences, as explained in Section 4.3.4. The transformer’s ability to process all elements in the input sequence concurrently allows for parallelized operations, essentially performing a ‘set-to-set’ transformation, resulting in an output sequence length that matches the input.

During training, we take the first 8 sequences of the output, which represent the predicted future listener. These are then compared to the ground truth listener data, as mentioned in Section 4.3.4. However, during test time, the model does not have access to any ground truth past listener motion data. Instead, it uses only the listener personality feature vector along with the speaker input to generate future listener motions.

5.2.2 Evaluation Metrics

In this section, we introduce a set of metrics to evaluate our model’s predictions, aligning with prior works [94, 11, 10]. Our evaluation encompasses the following:

1. **L2 Distance:** Comparing both listener sequences for ground truth y , and model prediction $\hat{y}$:

$$d_{L2} = \|y - \hat{y}\|_2 \quad (5.1)$$

2. **Motion Realism:** To assess the perceived realism of our generated listener motion sequences, we utilized the Frechet Inception Distance (FID) metric proposed by [106]. This metric measures the distributional similarity between the generated sequences and the actual ground-truth listener motion sequences present in our dataset. we calculate FID on the dimensional features space of the face (50 facial expression coefficients and 3D jaw euler angles), or upper body (306 parameters representing relative body joints) for the full motion sequences:

$$d_{\text{FID}} = \|\mu_{\hat{y}} - \mu_y\|_2^2 + \text{tr} \left(\Sigma_{\hat{y}} + \Sigma_y - 2(\Sigma_{\hat{y}}\Sigma_y)^{1/2} \right) \quad (5.2)$$

Where:

d_{FID} : distance score calculated on either face or upper body listener features.

$\mu_{\hat{y}}$: mean of the features of predicted listener motion sequences.

μ_y : mean of the features of ground truth listener motion sequences.

$\Sigma_{\hat{y}}$: covariance matrix of the features of predicted listener motion sequences.

Σ_y : covariance matrix of the features of ground truth listener motion sequences.

$(\Sigma_{\hat{y}}\Sigma_y)^{1/2}$: square root of the matrix product of covariance matrices.

$\text{tr}(\cdot)$: trace operator, sum of diagonal elements of a matrix.

3. **Motion Diversity:** This metric assesses the variability in motion within predicted listener sequence. We measure the variance across the time series of face or upper body motion sequences, specifically along the axis that represents the sequence’s temporal duration. Subsequently, we calculate the variance average over all sequences and face or upper body features.
4. **Listener-Speaker Correlation:** This metric assesses the quality of interactions between listeners and speakers by measuring distributional differences on pairs of listener-speaker sequences (P-FID). It is calculated in the same manner as the FID equation 5.2, but applied to concatenated listener and speaker (x) sequences of face or upper body features.

The P-FID distance is computed using the following equation:

$$d_{\text{P-FID}} = \|\mu_{y \oplus x} - \mu_{\hat{y} \oplus x}\|_2^2 + \text{tr} \left(\Sigma_{y \oplus x} + \Sigma_{\hat{y} \oplus x} - 2(\Sigma_{y \oplus x}\Sigma_{\hat{y} \oplus x})^{1/2} \right) \quad (5.3)$$

5.3 Quantitative Results

In this section, we present the quantitative evaluation of our final personality-aware model depicted in Figure 4-2, referred to as *Personality-Model II*, in the subsequent tables. This evaluation utilizes the metrics described in Section 5.2.2 to assess model performance. Specifically, we compare the performance of *Personality-Model II*, which

incorporates a refined approach to integrating personality traits, to *Personality-Model I*, which represents our initial approach to personality-aware modeling, and a *Base Model*, which does not account for personality traits.

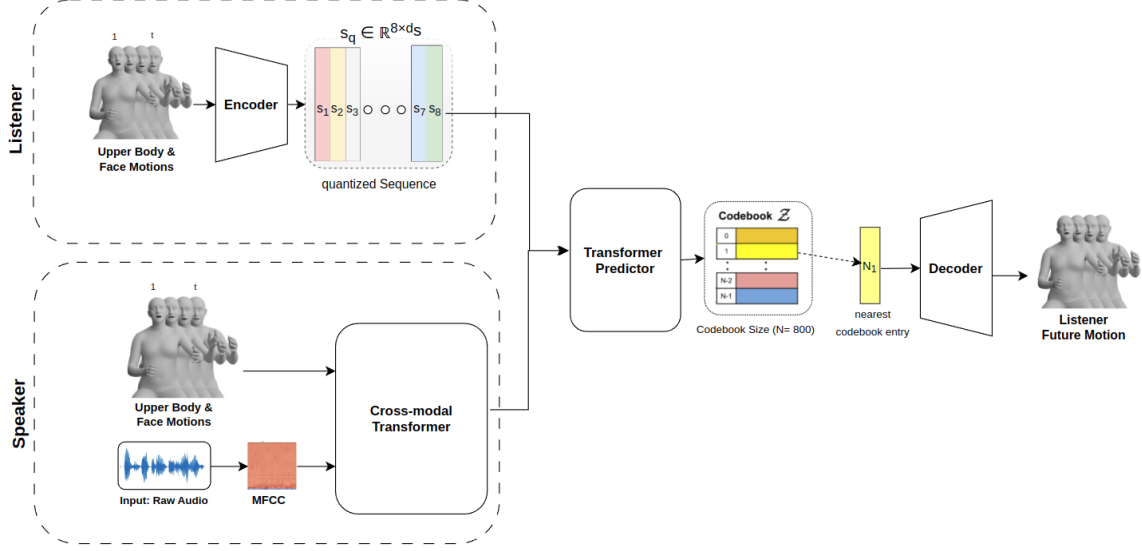


Figure 5-5: This modal serves as our baseline for generating listener motion responses without considering listener personality traits. It shares a similar architecture with the "Learning-to-Listen" model [11], and is trained on the face and upper body features extracted from the UDIVA dataset.

A. Base Model (Personality-Agnostic): This model, illustrated in Figure 5-5, serves as our baseline, focusing on generating listener motion response without the influence of listener personality traits. It is similar to the architecture presented in the "Learning-to-Listen" model [11]. However, it is trained on our extracted face and upper body features from the UDIVA dataset [13].

Implementation Details. Both the VQ-VAE and the Predictor are trained using the same optimizer, learning rates, codebook size, and training schedule as outlined in Section 5.2.1. Notably, the Personality Network 5.2.1(B.) is not utilized in this context since listener personality traits are not part of the input.

In Table 5.1, we present our results of our final personality-aware model *Personality-Model II* compared to personality-agnostic *Base Model*. The results show that *Personality-Model II* outperforms agnostic baselines on both facial and upper body motion prediction metrics. Notably, significant improvements are observed in the Fréchet Inception Distance (FID) and P-FID metrics for upper body motion, indicating more realistic listener responses and improved interaction quality with the speaker.

Model	Face (expression & jaw)				Upper Body (Body & hand)			
	L2 ↓	FID ↓	P-FID ↓	Var ↑	L2 ↓	FID ↓	P-FID ↓	Var ↑
Base-Model	32.45	7.67	10.47	1.39	75.29	58.87	96.82	0.97
Personality-Model I	33.05	8.65	11.94	1.49	74.41	51.90	91.49	0.84
Personality-Model II	32.12	6.15	10.31	1.54	72.26	43.16	87.73	1.03

Table 5.1: Presents a comparison of the performance of our final Personality-Aware Model "Personality-Model II" against the Personality-Agnostic "BASE Model", and also our initial "Personality-Model I" on the UDIVA dataset. The arrows next to each metric indicate the direction of improvement: (↓) signifies lower values are better, while (↑) indicates higher values are preferred. Bolded results denote significant improvements compared to the other models. The results show that Personality-Model II outperforms the other models on both Face and Upperbody metrics, with significant improvements in Upperbody FID and P-FID.

B. Personality-Model I: represents our initial attempt to incorporate personality traits into the model. In this approach, we fused the speaker’s motion, including both upper body and facial features, along with the audio without separating and processing face features and upper motion features independently as we did in our refined Personality-Model II illustrated in Figure 4-2. The architecture of the Personality-Model I is shown in Figure 5-6. We also employed a unified reconstruction loss for all features, rather than splitting them into sets of features as described in Section 4.3.1.1. However, this model performed worse than our refined Personality-Model II, as shown in Table 5.1.

Upon further analysis, we found that splitting the features into face and upper body features and using speaker face features with audio as input for the cross-modal transformer, then concatenating the output with the speaker’s upper body motion features, yielded better results. This improvement can be attributed to several factors. First, separating the features allows the model to focus on specific aspects of the input data, enabling more effective learning and feature extraction. Second, using speaker face features with audio as input for the cross-modal transformer helps the model to better capture the correlation between facial expressions and audio, leading to more nuanced listener responses.

C. Random Personality Inputs (for Personality-Model II): In this specific experiment, we evaluate the impact of the accuracy of personality trait inputs on the performance of *Personality-Model II*. To do this, we conduct a comparison using:

- i. **Actual Personality Scores:** Where *Personality-Model II* is provided with the genuine extraversion or personality score corresponding to each listener.
- ii. **Randomized Personality Scores:** Where all the corresponding listeners personality scores are randomized using the Python `random.shuffle`

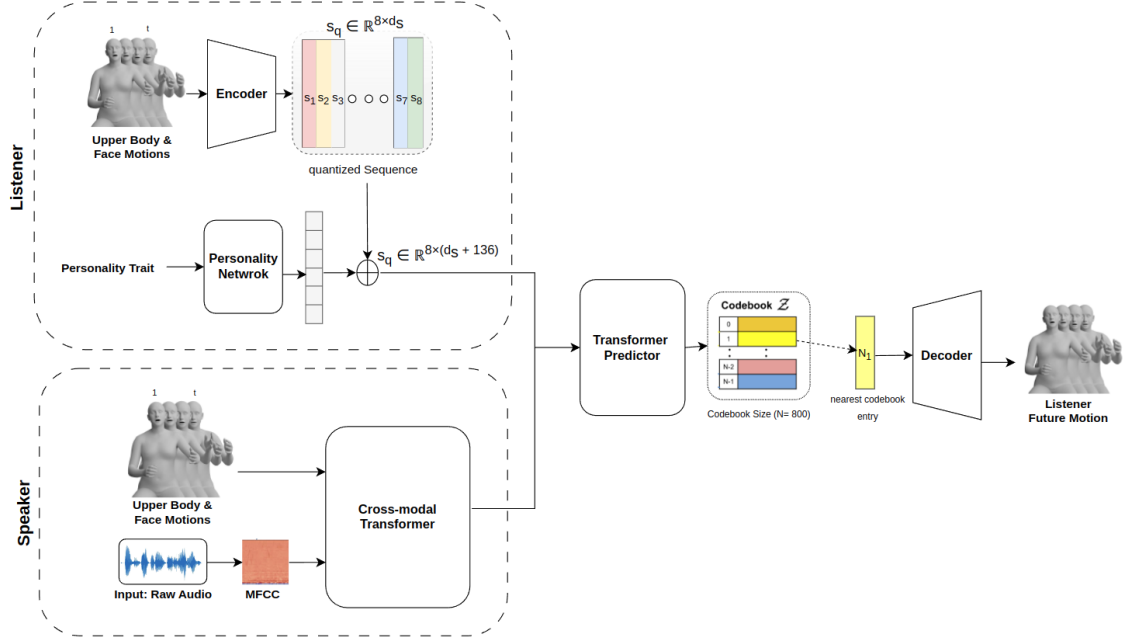


Figure 5-6: Personality-Model I Architecture: The figure shows the architecture of our initial attempt, Personality-Model I, where the speaker’s motion features are fused with audio without separating and processing face features and upper motion features independently.

function. This aims to simulate the scenario where the personality data input is unpredictable or inaccurate.

Input Type	Face (expression & jaw)				Upper Body (Body & hand)			
	L2 ↓	FD ↓	P-FD ↓	Var ↑	L2 ↓	FD ↓	P-FD ↓	Var ↑
Randomized Personality Scores	32.76	7.58	10.83	1.61	73.23	47.56	91.33	1.13
Actual Personality Scores	32.12	6.15	10.31	1.54	72.26	43.16	87.73	1.03

Table 5.2: Performance of *Personality-Model II* with Actual vs. Randomized Personality Inputs

This comparison assesses how the precision of input personality data influences *Personality-Model II* performance for generating listener motion sequences. The results of this comparison are illustrated in Table 5.2, dedicated to showcasing the performance of *Personality-Model II* under these two different input conditions. our bolded results in highlight significant improvements in model performance with actual personality scores, specifically observed in the FID and P-FID metrics for upper body features, underscoring the model’s enhanced ability to generate more realistic and personality-congruent non-verbal behaviors when informed by accurate personality traits. However, the variance values are slightly lower when the model is provided with actual personality scores compared to randomized inputs, indicating a reduction in the diversity of generated non-verbal behaviors. This suggests that while the model becomes more accurate and consistent in

reflecting the listener’s personality traits as reflect by FID and P-FD metrics, it may also produce a more uniform set of behaviors, potentially limiting the range of non-verbal cues represented.

5.4 User Study

In this section, we present two complementary user studies conducted as part of our research. These studies collectively explore the following key questions:

1. **Pre-training Evaluation (Study 1 5.4.1):** Before training our model on the PIXIE reference data extracted from UDIVA [13], we investigate whether participants can distinguish between different age groups (young or old), genders (male or female), and extraversion levels (extravert or introvert) based solely on listener non-verbal behavior, including facial expressions, body poses, and hand gestures.
2. **Post-training Evaluation (Study 2 5.4.2):** After training our final personality-aware model on the PIXIE reference data, we assess its capability to generate listener non-verbal behaviors conditioned on personality traits that remain distinguishable.
3. **Model Performance Comparison (Study 2 5.4.2):** Finally, by training both the personality-aware and personality-agnostic models on the same reference data, we aim to compare their performance and determine which model produces more interactive listener motion responses during speaker interaction.

In both of our studies, participants viewed a series of videos of 20 seconds duration. Each video presented a pair of 3D human listener avatars generated using PIXIE parameters. These 3D listener avatars depicted the upper body of the listener, including facial features and various body parts, such as arms and hands. Notably, we deliberately chose a neutral 3D listener avatars by setting all PIXIE shape coefficients to zero. This deliberate choice aimed to ensure that participants evaluate the listener’s non-verbal behavior exclusively, focusing only on facial expressions and body gestures, without any gender bias. After each video, participants were asked a question and given sometime to respond.

5.4.1 Study 1: Assessing Listener Non-Verbal Traits

We conducted the experiments for this study via Google Forms. Our aim is to assess participants ability to distinguish various non-verbal behavior traits displayed by listeners, including age, gender, and extraversion. Specifically, we sought to determine whether participants could accurately perceive these distinctions solely through visual cues from the listener’s non-verbal behaviors, such as facial expressions, body posture, and hand gestures. The experiments are outlined as follows:

- **Age Trait Comparison:** We presented pairs of 3D listener avatars in a series of videos, with each pair representing two distinct age groups (young and old). After each video, we ask each participant " Which of the two listener avatars is older? LEFT or RIGHT". To create these pairs, we randomly selected 6 listeners for each age group (old/young) from the top 20% and bottom 20% age quantiles of the training set. We ensured an equal gender distribution in both groups and similar extraversion averages. For each listener in the young age group, we selected a counterpart from the old age group with the same gender, and closely matching the extraversion score of the old age group listener.
- **Gender Trait Comparison:** We displayed videos of 3D listener avatars, comparing genders (male and female), and participants had to chose " Which of the two listener avatars is male? LEFT or RIGHT". We ensured that for each video, one listener was male, and the other was female. We achieved this by randomly selecting 6 male listeners and 6 female listeners, all with similar age and extraversion score averages, to ensure that we are only comparing with respect to the gender trait only. For each male listener, we compare with a female listener with close age and extraversion score.
- **Extraversion Trait Comparison:** : We showcased videos of listener avatars displaying high and low extraversion. Participants were ask to identify "Which listener avatar exhibited more extraverted behavior? LEFT or RIGHT". For this comparison, we randomly selected 6 listeners for each group (high and low extraversion), with an equal distribution of 3 females and 3 males in each group, and with similar age averages.

Extraversion, as defined in this study, is a broad personality trait that encompasses a number of more specific characteristics such as sociability, assertiveness, high activity level, positive emotions, and impulsivity. [26]

Additionally, the left-right position of the two listener avatars in each video was randomly shuffled to ensure that participants' responses were not influenced by the position of the avatars in the videos.

Key Findings: In our study, which involved 20 participants (11 males, 8 females, and 1 who preferred not to say), participants were able to identify extraversion traits with a 66% success rate (means that two-thirds of the time, participants were able to correctly discern whether an listener avatar displayed extraverted traits), age traits with a 51% success rate, and gender traits with a 48% success rate. The proficiency in recognizing extraversion traits was notably higher than that for age and gender. Figure 5-7 shows a pie-chart illustrating the distribution of success rates for identified traits among participants.

Building on our initial study findings, where extraversion traits were more accurately identified than age and gender. We initiated a second study focused on evaluating our final model ability to produce non-verbal listener behaviors reflective

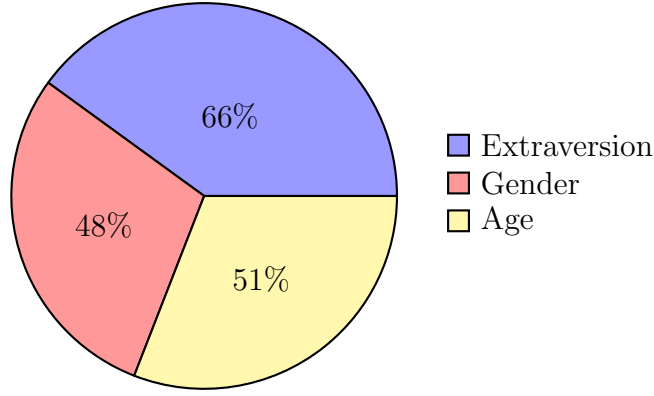


Figure 5-7: Study 1 results: showing the success rate of each trait of Gender, Age, and Extraversion. With two-thirds of the time, the 20 participants were able to correctly discern whether an listener avatar displayed extraverted traits.

of different extraversion levels. This decision was informed by the distinct recognizability of extraversion traits observed in our preliminary data. Next, we will detail the methodology and outcomes of this follow up study.

5.4.2 Study 2: Evaluating Final Model Predictions

In this phase, we aimed to evaluate the capabilities of our final models, specifically after training it to recognize listener behaviors conditioned solely on extraversion personality traits. Conducted using PsychoPy Software¹, this study comprised two key experiments:

- Model’s Ability to Generate Distinguishable Behavior:** In the first part of the study, we evaluated our final personality-aware model ability to generate listener avatars conditioned on different levels of extraversion personality traits. Specifically, we utilized the highest and lowest extraversion scores from our dataset as conditions to generate two distinct listener avatars. We then presented these avatars side by side in a series of 20-second videos alongside the speaker, and asked participants ”Which listener avatar exhibited more extraverted behavior? LEFT or RIGHT”. This experiment aimed to assess whether our model could produce distinguishable non-verbal behaviors for listeners with varying extraversion levels.
- Comparison of Personality-Aware and Personality-Agnostic Models:** In the second part of the study, we sought to determine whether our final personality-aware model, which considers personality traits during generation,

¹PsychoPy is an open-source application allowing for the execution of neuroscience, psychology, and psychophysics experiments. It is designed for the creation of precise, timing-accurate experimental stimuli and the collection of data. See Peirce et al. [107]

outperformed the personality-agnostic model. We presented participants with videos featuring two listener avatars side by side with the speaker, one of the listener avatars generated by the personality-aware model using actual extraversion scores as input and the other by the personality-agnostic model. After each video, participants were asked to identify "Which listener avatar seems more listening and more engaged with the speaker? LEFT or RIGHT". This experiment aimed to compare the performance of both models in generating interactive listener motion responses during speaker interaction.

It's important to clarify that our evaluations and experiments for this study were conducted on the test set. This approach ensures that we assess our model's performance on listeners it has never encountered before, offering a reliable measure of its capabilities and maintaining consistency with our quantitative results. Specifically, we randomly selected 6 listeners from the test set for each experiment.

In the first experiment, we employed the highest and lowest extraversion scores from our dataset to condition each of the 6 listeners. This methodology allowed us to utilize our personality-aware model to generate two distinct avatars for each listener, one for each of the extremity scores. Moreover, we maintained consistency with the initial study by adopting the same definition of extraversion [26].

For the second experiment, we used each of the 6 randomly selected listeners as inputs for two distinct models: one generated an listener avatar based on actual extraversion scores through a personality-aware model, while the other was produced by a personality-agnostic model, which does not incorporate personality traits. The aim was to conduct a side-by-side comparison of the models outputs to showcase the effect of integrating personality traits into the listener generation versus omitting them.

Similar to the first study, we randomized the positions of the two listener avatars in each video. This was done to prevent any bias that could influence participants responses based on the listener avatars locations in the videos.

Key Findings: The study involved 20 participants including 12 females and 8 males to evaluate both models, revealing that *Personality-Model II*, a personality-aware model, is capable of accurately differentiating behaviors between introverted and extroverted listeners, with 86% of the time, participants recognizing these distinctions. Moreover, 71% of the time, participants showed a preference for *Personality-Model II* over the personality-agnostic *Base Model*, specifically when *Personality-Model II* utilized actual extraversion scores. This comparison underscores the enhanced performance of *Personality-Model II* to reflect listener non-verbal behavior responses by utilizing actual extraversion scores (refer to Figure 5-8)

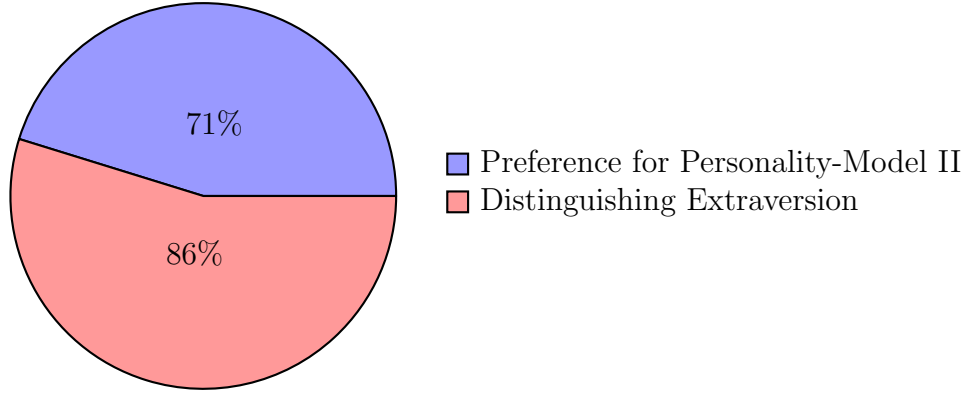


Figure 5-8: Study 2 results: Evaluation of *Personality-Model II* showed 86% of the time participants can distinguish between introverted and extroverted behaviors, and with 71% preference over the *Base Model* when utilizing actual extraversion scores.

5.5 Qualitative Results

In this section, we present a qualitative analysis of our personality-aware model’s ability to differentiate between introverted and extroverted listener behaviors. Figure 5-9 offers a glimpse into a sequence of 6 frames from our second user study, as discussed in Section 5.4.2. The figure emphasizes the model’s proficiency in capturing subtle non-verbal cues and reveals distinct differences in engagement and expression between introverted and extroverted listener responses.

In Figure 5-9, we compare the speaker’s frame sequence to both the introverted listener avatar sequence (first row) and the extroverted avatar sequence (second row). The introverted listener avatar in the first row displays minimal facial expressions, such as not smiling when the speaker does, and tends to avoid eye contact or looking towards the speaker. Additionally, the introverted listener exhibits limited body and hand gestures.

In contrast, the extroverted listener in the second row is seen mostly smiling and displaying a tendency to lean and gaze towards the speaker, indicating greater engagement. The extroverted listener also exhibits more active body and hand gestures, such as touching the face while looking at the speaker, appearing to listen carefully, and reacting to the speaker. This visual comparison highlights the model’s ability to differentiate listener responses based on personality traits, with extroverted listener demonstrating more pronounced engagement behaviors.

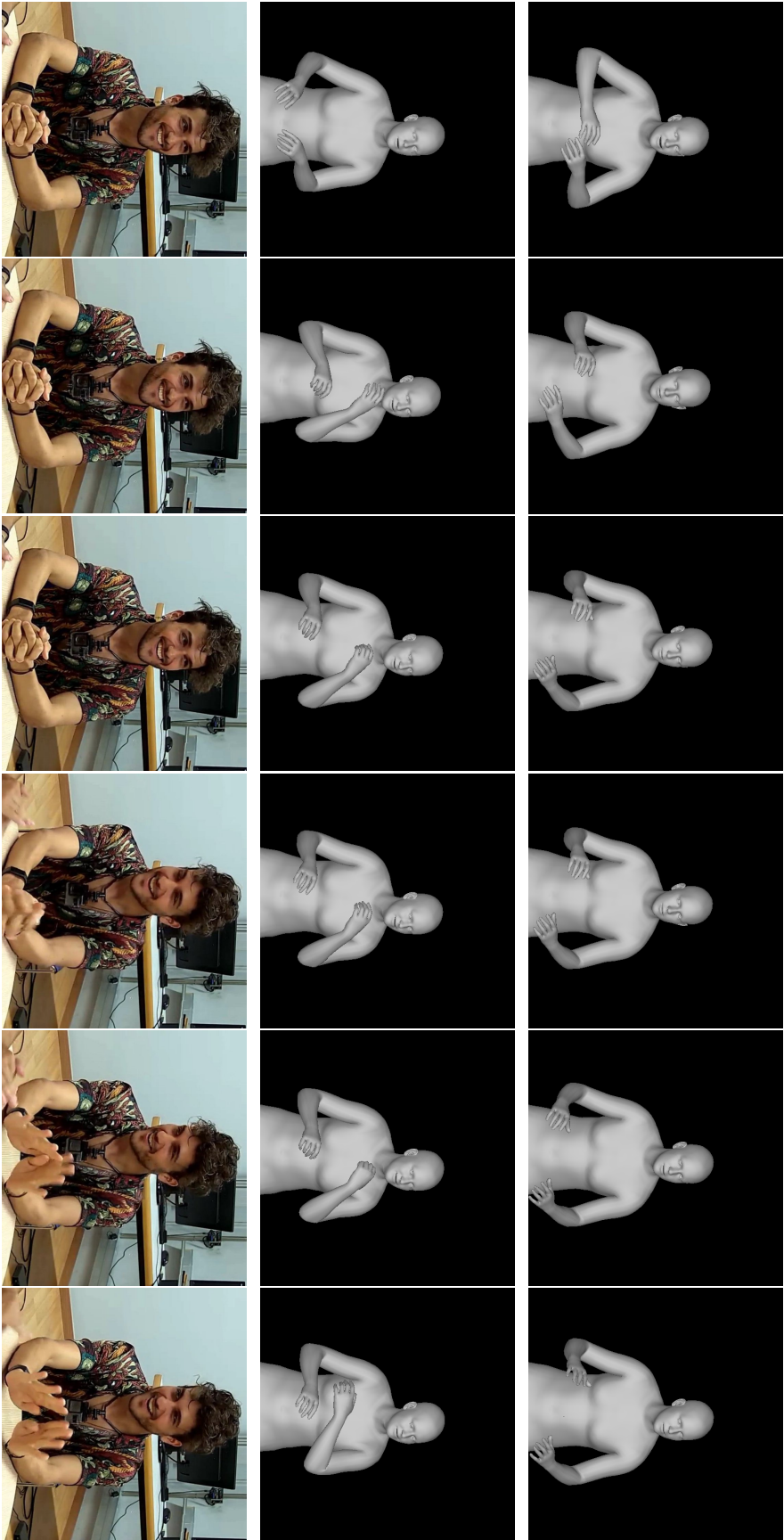


Figure 5-9: Frames sequence from our second user study demonstrating the model’s ability to generate distinct listener behaviors based on personality traits. Introvert and extrovert listener avatars are shown in the first and second rows, respectively. The speaker’s frame sequence is provided in the last row for reference.

Chapter 6

Discussion, and Future Work

In this chapter, we present our key findings, explore their potential applications, and discuss the limitations of our approach. We will also suggest directions for future research in this area.

6.1 Discussion

6.1.1 Motion Features Analysis

In our analysis, we have identified limitations associated with using PIXIE for extracting motion features from video sequences. While PIXIE excels in reconstructing detailed human models from single images, its application to video data presents challenges. Specifically, since PIXIE processes each frame independently, it can lead to jittering in motion sequences, detracting from the natural flow of movements.

To address this, we implemented a mean average filter with a window size of 7 on the PIXIE parameters. This filter smooths the motion sequences, reducing jitter. However, this approach has a drawback: over-smoothing can result in a loss of motion detail, potentially diminishing the expressiveness of the listener behavior. Balancing the smoothing effect to preserve motion variability while minimizing jitter is a key concern.

Moreover, PIXIE relies on a single viewpoint for reconstructing the 3D human body from each frame. This approach has limitations when dealing with occlusions, which occur when parts of the body are hidden from view (e.g., hands under a table). Since PIXIE only sees what’s visible from its single perspective, it can’t accurately infer the pose or motion of occluded body parts. This leads sometimes to reconstruction errors and misalignments in the extracted motion features.

These challenges highlight areas for future improvement in motion feature extraction and underscore the need for careful consideration of the methods used to ensure they contribute positively to model performance.

6.1.2 Metrics Analysis

In this section, we compare the performance of our best personality-aware model, referred to as "Personality-Model II" (shown in Tables 5.1 and 5.2), trained on features extracted from the UIDVA dataset as described in Section 5.1, with that of the personality-agnostic Baseline model (labeled "Base-Model" in Table 5.1). We assess both models across the metrics outlined in Section 5.2.2, utilizing the entire test set comprising 15 previously unseen UDIVA listeners or participants to ensure a fair comparison.

The results demonstrate that *Personality-Model II*, by incorporating listener personality extraversion trait, outperforms the *Base-Model*, which does not include training on such trait, across various performance metrics. Although there is a slight improvement in the facial metrics, our upper body metrics exhibit significant enhancements. More specifically, in both motion realism metrics FID and P-FID, which translates to more realistic listener responses and improved interaction quality with the speaker.

Interestingly, while *Personality-Model II* demonstrates a improved performance across most metrics when actual personality scores are used, it exhibits a lower variance in motion diversity compared to the randomized personality inputs as shown in Table 5.2. This suggests that the model, when accurately informed by the listener personality, particularly extraversion, generates more consistent, or uniform motion sequences and possibly less varied motion sequences. Although this might imply a decrease in the richness of generated listener non-verbal behaviors, it also reflects the model ability to generate motions closely aligned with specific personality traits. This leads to a trade-off between personalizing listener responses and preserving a wide range of motion diversity. This finding prompts further exploration to maintain this balance enhancing realistic, personality-consistent non-verbal listener behaviors without compromising motion diversity, marking a vital aspect for future research.

6.1.3 User Study Analysis

Our research comprised two distinct, yet complementary, user studies designed to assess the effectiveness and nuanced performance of our models in generating listener non-verbal behaviors, with a specific focus on the role of personality traits, notably extraversion. The first study, a pre-training evaluation, involved 20 participants (11 males, 8 females, and 1 who preferred not to say) and aimed to determine the ability of participants to discern between various non-verbal behavior traits including age, gender, and extraversion, without any influence from our model. The findings from this initial study highlighted a notable proficiency among participants in recognizing extraversion traits with a success rate of 66%, suggesting a clear visibility of such traits through non-verbal cues alone. This success rate was significantly higher compared to the ability to identify age (51%) and gender traits (48%), indicating a more pronounced recognition of personality over other demographic characteristics.

Building on these insights, the second study was conducted post-training with the aim of evaluating our final personality-aware model, *Personality-Model II*, against a personality-agnostic baseline, *Base-Model*. This study also engaged 20 participants, achieving a balanced gender distribution of 12 females and 8 males. The dual experiments within this study were specifically designed to assess (1) the model’s ability to generate distinguishable listener behaviors based on extraversion levels, and (2) to compare the effectiveness of the personality-aware model against the personality-agnostic model in producing interactive and engaging listener responses.

The key findings from the second study are particularly illuminating. *Personality-Model II* demonstrated a significant capability in generating behaviors that participants could distinguish between introverted and extroverted listeners with an 86% success rate. This not only affirms the model’s effectiveness in capturing and replicating the nuanced expressions of extraversion through non-verbal cues but also validates our hypothesis that personality traits, especially extraversion, can be successfully encoded into listener responses.

Furthermore, when comparing the performance of *Personality-Model II* with the *Base Model*, participants exhibited a marked preference for the personality-aware model, with 71% favoring it over the personality-agnostic counterpart. This preference underscores the added value of incorporating personality traits into the model, enhancing the realism and engagement of the listener avatars responses. This finding is particularly noteworthy as it aligns with our metrics analysis from the previous section, where *Personality-Model II* outperformed the *Base-Model* across various performance metrics, notably in motion realism and listener-speaker interaction quality as indicated by improvements in facial and upper body metrics.

Our findings from the second study, as elaborated in the qualitative Section 5.5 of the evaluation chapter, reveal that extraverted listeners showcased specific non-verbal behavior patterns, such as leaning forward towards the speaker, displaying energetic or expressive behaviors, engaging in more interactive facial expressions, and using body and hand gestures more frequently. These observed behaviors align with existing research, which reports that extraversion influences certain non-verbal behaviors including leaning the upper body forward, tilting the head, energetic physical movement, and higher rates of eye contact and gestures [24, 29, 30, 25, 31].

6.2 Applications

Our personality-aware listener model introduces exciting possibilities for towards a more personalized interaction:

- **Personalized Learning:** Imagine intelligent tutoring systems that adjust to a student’s learning style and personality. For introverts, a calmer tutor with minimal gestures; for extroverts, a more energetic guide. This fosters a more engaging learning experience.

- **Emotionally Intelligent Virtual Agents:** Customer service encounters can benefit from personality-aware models. Imagine virtual agents that adapt to customers’ personalities. Extroverted customers might get a more enthusiastic agent, while introverts encounter a calmer one. This personal touch can lead to better customer experiences and conflict resolution.
- **Enhanced Social Connection with Telepresence Robots:** Telepresence robots can leverage personality-aware models to create more natural interactions. The robot can tailor its non-verbal cues (gaze, posture) to the user’s personality, fostering a stronger sense of connection and personalized care, especially in caregiving settings. This personal touch can help combat social isolation.

6.3 Limitations and Future Work

In this thesis, we have examined the development of a personalized listener avatar for dyadic interactions, with an emphasis on generating responses that reflect both the actions of the speaker and the personality traits of the listener. Despite the progress made, there remain several unresolved challenges that offer directions for future work.

- **Dual Response Modeling:** Our method, like prior works [11, 34, 35, 36, 5], has primarily modeled the listener’s response to the speaker. Future work could aim to generate full conversational responses [108] for both the speaker and the listener, aligning more closely with real-life dyadic interactions. This comprehensive approach would significantly enhance the realism and dynamism of simulated conversations.
- **Motion Feature Extraction Limitations:** The reliance on motion features extracted using PIXIE, which is constrained by its single-viewpoint approach, presents a limitation as been discussed in Section 6.1.2. This method can lead to inaccuracies in reconstructing occluded body parts, such as hands beneath a table, causing misalignments in motion features. Future research should consider occlusion and viewpoint diversity in motion feature extraction to achieve more accurate and realistic human reconstructions [109].
- **Incorporation of Semantic Speech Content:** Currently, our model does not account for the semantic content of the speaker’s speech, relying solely on encoded audio frequencies. However, the listener’s responses can be significantly influenced by the content and meaning of what is being said. Integrating large language models to parse and understand the semantic content of speech presents a promising area for future development, potentially enabling more nuanced and context-aware listener responses [110].
- **Beyond Extraversion:** While our model accounts for the listener’s extraversion, listener non-verbal behavior is influenced by a wider range of factors, including educational background, cultural context, and other Big Five personality

traits, including Agreeableness, which is linked to increased head nodding [24]. Future iterations of our model could benefit from incorporating these additional dimensions to provide a more comprehensive simulation of non-verbal behavior.

- **Trade-off Between Personalization and Motion Diversity:** Our findings indicate that while the incorporation of actual personality scores enhances model performance across most metrics, it leads to a reduction in motion diversity as seen in Section 6.1.2. This suggests a trade-off between personalizing listener responses and preserving a wide range of motion diversity. Future research should explore strategies to balance these aspects, ensuring that the enhancement of realistic, personality-consistent non-verbal behaviors does not compromise the richness of generated motions.

Chapter 7

Conclusions

In this thesis, we presented a novel method for generating personality-aware listener avatar in dyadic interactions. Our method builds on the transformer-based model from "Learning to Listen" by Ng et al. [11], and expands it to include facial expression, upper body, and hand movements that reflect the listener's extraversion trait. By incorporating listener personality traits into the model, we aimed to improve the realism and engagement of listener responses to the speaker.

To evaluate our approach, we conducted two user studies. The first study assessed the ability of participants to recognize various traits based on listener non-verbal behavior, including extraversion, gender, and age. The results showed that participants were able to identify extraversion traits with a 66% success rate, making it the most distinguishable trait among the ones tested. Building on these findings, we initiated a second study focused on evaluating our final model's ability to produce non-verbal listener behaviors reflective of different extraversion levels. This decision was informed by the distinct recognizability of extraversion traits observed in our preliminary data.

Revisiting our research questions from Section 1.2, the first question was "Is it feasible to build a model that generates non-verbal behaviors that encompass the entire upper body, including facial expressions, arms, hands, and body posture, and that are reflective of the listener's personality trait?" The results of the second user study demonstrated that our personality-aware model can indeed generate distinguishable non-verbal behaviors based on extraversion levels. Participants showed a clear preference for the extraverted listener avatar generated by our model, correctly identifying it in 86% of cases.

The second research question was "Can the incorporation of listener personality traits into the model improve the realism and engagement of listener avatars responses to the speaker?" The second user study also showed that our personality-aware model produced more interactive, and engaging listener motion responses compared to a personality-agnostic model. This improvement was observed in both the user study results, and the quantitative results on various metrics.

The findings of this research have potential applications in many areas such as virtual reality, gaming, and social robots, that enhance user experiences. However, there are also limitations to this research. One limitation is that we only focused on the listener’s personality trait of extraversion, and future work could explore the incorporation of other personality traits into the model. Additionally, our method primarily models the listener’s response to the speaker, and future work could aim to generate full conversational responses for both parties to enhance the realism and dynamism of simulated conversations.

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